

Identifying Panel Conditioning with Refreshment Samples: Sharp Bounds and Design Assumptions*

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September 23, 2026

Abstract

Refreshment samples are the standard remedy for panel attrition, and the identification results behind them maintain that participation does not change measurement. We characterize what a refreshment sample identifies about panel conditioning, modelled as a deterministic monotone map at reinterview, when attrition is unrestricted. A candidate map is consistent with the data if and only if the retention-scaled distribution of the stayers' implied latent outcomes is setwise dominated by the refreshment distribution; every such map is rationalized by an explicit attrition process. Without attrition the map is point identified; with attrition, a density-ratio condition governs the identified set, and tail behaviour alone does not determine it. For an unrestricted map the survivors' mean effect has the Horowitz–Manski trimming bounds; for an item with all categories reported, the model reduces to a test of no conditioning. Within a cohort, other waves, dropout patterns and entry-wave items leave the set unchanged unless restrictions link selection across waves. We state survival matching, a symmetric variant and the entry-wave functional of Das, Toepoel and van Soest as such restrictions, derive their biases, and give rank conditions under which refreshment schedules identify the curvature of the conditioning path. A Japanese panel illustrates the results.

Keywords: panel conditioning; refreshment samples; attrition; partial identification; survey design; rotation panels

*Code for every simulation and deterministic check in this paper is in the replication archive at <https://github.com/sokubo/paper-refreshment-designs-replication> (fixed version: tag `paper-v0.5`, commit `0869ad8`). The empirical illustration uses licensed JLPS microdata, which are not redistributed; the archive contains the code that produces the reported aggregates and a synthetic example (see Data and code availability). This work was supported by JSPS KAKENHI Grant Number 22K13525.

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1 Introduction

Research on refreshment samples and research on panel conditioning impose different restrictions on selection and measurement. The literature on attrition in panel surveys treats refreshment samples as the key to identification: a fresh cross-section drawn at a later wave reveals the population distribution that the retained sample no longer represents, and under additively separable models of nonignorable attrition the joint distribution can be recovered (Hirano et al. 2001; Deng et al. 2013). These models maintain that survey participation does not alter measurement. In Hirano et al. (2001) the restriction is carried by the refreshment sample's status as a fresh draw from the same population; the paper discusses repeated response and the possibility that effort during a first interview affects later response (Hirano et al. 2001, 1646, 1648), but its attrition model does not explicitly model changes in measurement caused by prior interviews. Deng et al. (2013) discuss panel conditioning as a threat to the design but do not incorporate it, since their additive-nonignorable model restricts the attrition probability rather than the measurement. The literature on panel conditioning documents that participation can alter reports (Warren and Halpern-Manners 2012; Bailer 1975) and treats attrition as a threat to its own comparisons: Warren and Halpern-Manners (2012, 508–10, 517) discuss attrition and mode as threats and describe comparisons of old and new cohorts restricted to two-wave survivors, and Halpern-Manners, Warren, and Torche (2017, 108–9 and note 14, p. 119) explain a survival-matching comparison, the threat of attrition processes that differ by cohort, and a regression diagnostic for it. Das, Toepoel, and Soest (2011) bound conditioning effects for binary items under worst-case attrition and discuss the relation of their model to that of Hirano et al. (2001). Official-statistics work on rotation group bias documents systematic differences by time in sample without being able to say how much of them is measurement: Krueger, Mas, and Niu (2017, 260, Table 1, and p. 262) find that adjusting for nonresponse accounts for more than one-third of the rise in rotation group bias after 1993, caution that their observational evidence cannot establish causation unambiguously, and suggest (p. 264) that the bias need not be an inherent consequence of repeated interviewing. What we have not found written down is the sharp identified set of a conditioning map when attrition is unrestricted, or a statement of the common designs'

comparability requirements as restrictions on latent outcome distributions or means.

This paper characterizes identification in a deterministic monotone measurement model with unrestricted attrition. Given a panel with nonignorable attrition *and* panel conditioning, what does a refreshment sample identify about the conditioning map? A candidate map is consistent with the data if and only if the retention-scaled distribution of the stayers' implied latent outcomes is setwise dominated by the refreshment distribution (Theorem 1); the set this defines is sharp, with an explicit construction of an attrition process that rationalizes any point in it.

The characterization yields several implications for identification and survey design. First, it isolates where identification comes from: the constraint uses only the stayers' reported outcome distribution, the refreshment distribution, and the retention rate. The stayers' joint distribution with their baseline reports is not needed to identify the conditioning map; it matters only for reconstructing a selection process afterwards. Second, it describes the role of attrition. With full retention the constraint collapses to an equality that point-identifies the map as a quantile alignment, with no assumption on selection; as retention falls, the constraint relaxes into a set of observationally equivalent maps. For location shifts that set is governed by the ratio of the selection-weighted latent density to the refreshment density (Corollary 2). Tails constrain it—when selection is bounded away from zero, a refreshment density with tails thinner than exponential excludes every perturbation that moves the implied latent distribution towards such a tail—but do not by themselves determine it, because the constraint must hold at every outcome value and can bind in the centre of the distribution. Third, it relates to known bounds. Das, Toepoel, and Soest (2011) derive sharp worst-case bounds on the conditioning effect for a binary item, of width equal to the attrition rate; their model lets individual binary responses change at reinterview, whereas a deterministic increasing map on a two-point support can only fix both points, so the two models are compared rather than nested. For the survivors' mean effect under an unrestricted map, the sharp bounds are the trimming bounds of Horowitz and Manski (1995) and Lee (2009) (Corollary 6)—.70 standard deviations wide at 80% retention for a Gaussian refreshment distribution—and a location restriction narrows them. Fourth, it covers attri-

tion that does not depend on the latent outcome: the identified set for a location shift is then the sublevel set of the refreshment density’s shift-ratio function at the inverse retention rate (Corollary 3)—a single point for the Gaussian and for any density with tails thinner than exponential, and an interval of half-width $\lambda^{-1} \log(1/p)$ for the Laplace and logistic densities—so that what identifies conditioning in that case is a property of the population’s distribution rather than the benignity of the attrition, and a plug-in width computed on a trimmed support measures the trim rather than the selection (Remark 2). Fifth, it separates continuous from discrete items: for an item all of whose categories are reported by the stayers, the only admissible deterministic map is the identity, so the theorem reduces to a test of “no conditioning” that holds under every attrition process, and conditioning found on such items has to be modelled as a stochastic change of answers (Corollary 4).

Within a cohort, the longitudinal structure of a panel does not change this unless restrictions link the waves. One might hope that the joint distribution of a cohort’s reports over many waves, the timing of its dropouts, or a battery of items asked once at entry would narrow the identified set of the map at a given wave. Theorem 2 shows that, without restrictions linking selection across waves, they do not: the identified set of the map at a wave with a refreshment sample is exactly the two-wave set computed from that wave’s respondents and its refreshment distribution, the maps at other waves are unidentified, and every observed pattern of participation is reproduced by an explicit multiwave completion. Within the maintained model, additional identifying restrictions must therefore exclude otherwise feasible completions, and the necessity results below are statements about the full distribution of the data rather than about a few moments.

The second half of the paper turns from delimitation to recovery. A single refreshment sample set-identifies conditioning. Two refreshment samples at different waves, generating several tenure configurations in common periods, together with an entry-wave negative-control battery and a transport restriction on selection with a known loading, point-identify the conditioning contrasts at the tenures observed at refreshment episodes (Theorem 4); a bounded loading gives a set. The transport restriction is an assumption, not a consequence of overlapping birth years. We distinguish two kinds of refreshment: *age-matched* refreshments, whose

birth years overlap those of the incumbent cohorts (in the Japanese panel of Section 10, the 2011 refreshment), and *generational* refreshments, which recruit a new birth cohort with no overlap (the 2019 refreshment)—for which levels are only interval-identified under a bound on residual cohort effects, while increments between incumbent tenures remain identified under the same restrictions (Corollary 7). A companion empirical paper implements the single-refreshment diagnosis; this paper supplies its identification theory and a design theory for panels that want more.

The third part is constructive. Applied work combines a single refreshment sample with a restriction on attrition, often stated in words, to obtain a point estimate; the survival-matching comparison of Halpern-Manners, Warren, and Torche (2017) is the best known. Section 7 expresses the comparability requirements of such designs as explicit restrictions on latent outcome means. Each is a restriction on the *selection functional*, the survivors’ advantage on the latent outcome, that identifies the survivors’ mean shift: survival matching reads the survivors’ latent mean off the fresh cohort’s future, a symmetric variant does so after equalizing the number of selection decisions, and an entry-wave correction reads it off the continuing cohort’s own past, using the fact that entry reports are unconditioned by construction. The entry-wave functional is not new: it is the observable combination that Das, Toepoel, and Soest (2011, 43–44, Assumption 3) use under their stationary-attrition-bias assumption. We restate it as a restriction on the selection functional, which shows that it identifies a *survivor* effect under a restriction on *unconditioned* latent outcomes, whereas their assumption identifies a population effect under a restriction on conditioned ones—a distinction that matters when effects are heterogeneous—and we add the symmetric variant. We give exact bias identities for each design and, under an explicit model with person-level traits, independent transient innovations and per-wave response decisions, which designs are biased under state-dependent and under non-stationary attrition, and the direction of the bias under state dependence (Proposition 5); the disagreements between designs then assess the compatibility of the maintained restrictions without uniquely identifying a failure mechanism (Section 8). Section 9 treats the schedule itself as the object of choice. A rank condition states, before data arrive, which functionals of the conditioning path a proposed schedule identifies; the *stride* of the schedule (the greatest common divisor of the spacings

between entry cohorts) fixes a set of directions that are never identified, so that a panel refreshed at a common interval greater than one cannot identify ordinary curvature, and an entry at a coprime spacing, followed for long enough, can restore it (Theorem 5, Proposition 7). A balance–information result restates, for published aggregates, the classical finding that tenure-balanced rotation keeps time-in-sample bias constant in levels and absent from changes (Bailar 1975; Park, Kim, and Choi 2001), and adds that when only the balanced aggregate is released, the published series depends on the time-in-sample path only through its weighted average, which is itself confounded with the period level, so no functional of the path is identified from it (Theorem 6).

Throughout, “conditioning” is a measurement map applied to latent outcomes at reinterview, in the sense of our companion paper on the identification of time-in-sample effects (the *companion identification paper* below); the present paper takes the two-wave measurement problem as primitive and asks what cross-sectional anchors buy. Related formal work includes Feng, Hu, and Sun (2022), who restrict response dynamics to first-order dependence on the previous report, which makes entry-period measurement error depend on the current true status alone—because a first interview has no previous report to condition on—and combine this with rank and eigenvalue conditions to point-identify misclassification probabilities in the Current Population Survey; Franguridi and Kosenkova (2026), who study refreshment identification of the attrition process without conditioning; Chadi (2021), who uses the coexistence of entrants induced by the data collector and “natural” entrants into a household panel to separate attrition bias from participation-experience effects under regression-based identifying restrictions, including experience effects common to both types of entrant, without characterizing the sharp set for an unrestricted conditioning map; Bertoli, Jakli, and Pascoe (2026), who formalize conditioning bias in two-wave event-impact surveys and propose a dual randomized survey design that avoids re-asking identical items; and Okubo (2024), who states potential-outcome estimands and identification assumptions for conditioning with a refreshment sample. The contribution here is the sharp identified set under both frictions in the deterministic monotone model, its multiwave extension, and a formal statement of the design restrictions, their biases, and the rank conditions of refreshment schedules.

2 Setup and Observables

A cohort enters at wave 1 with latent outcomes (Y_1, Y_2^*) jointly distributed as F on $\mathbb{R}^2$, with marginals F_1 and F_2 . Every cohort member reports Y_1 at wave 1. Between waves, attrition selects stayers: $S \in \{0, 1\}$ with selection function

$$\pi(y_1, y_2) = \Pr(S = 1 \mid Y_1 = y_1, Y_2^* = y_2), \quad p = \Pr(S = 1) = \int \pi dF > 0,$$

unrestricted—attrition may depend on the latent wave-2 outcome itself (nonignorable). At wave 2, stayers report a *conditioned* measurement

$$Y_2 = c(Y_2^*),$$

where c belongs to $\mathcal{M}$, the class of increasing homeomorphisms of $\mathbb{R}$ (continuous, strictly increasing, and onto, so that c^{-1} is defined on all of $\mathbb{R}$). (Location shift $c(y) = y + h$ is the leading parametric case.) A refreshment sample drawn at wave 2 from the same population reports Y_2^* without conditioning (“refreshment validity”).

The observables are the tuple

$$\mathcal{O} = (F_1, F_2, G, p),$$

where G is the joint distribution of (Y_1, Y_2) among stayers, with first marginal G_1 and second (reported-outcome) marginal Q . All four components are directly estimable: F_1 from the full entry cohort, F_2 from the refreshment sample, G and p from the panel’s retention records. Throughout, distributions on $\mathbb{R}$ are assumed atomless where quantile alignments are invoked.

A *structure* is a triple $\theta = (F, \pi, c)$ with F a distribution on $\mathbb{R}^2$, $\pi : \mathbb{R}^2 \rightarrow [0, 1]$ measurable, and $c \in \mathcal{M}$. Structure θ *rationalizes* $\mathcal{O}$ if it reproduces all four components. The identified set for the conditioning map is

$$\mathcal{C}_{ID} = \{c' \in \mathcal{M} : \exists (F', \pi') \text{ such that } (F', \pi', c') \text{ rationalizes } \mathcal{O}\}.$$

Two pieces of notation. For $c' \in \mathcal{M}$, write $Q_{c'} := (c'^{-1})_{\#}Q$ for the pullback of the stayers’

reported-outcome distribution—the distribution the stayers’ *latent* Y_2^* must have had if the conditioning map were c' . For measures μ, ν write $\mu \leq \nu$ for setwise domination: $\mu(A) \leq \nu(A)$ for every Borel A (equivalently $d\mu/d\nu \leq 1$ where densities exist).

3 The Main Theorem

The engine is a completion lemma showing that any candidate stayer sub-distribution consistent with the marginals can be embedded in a full structure. It is an instance of the existence results for measures with given marginals of Strassen (1965, Theorems 6–7, pp. 432–434), which include domination constraints; we use the explicit residual-product construction below rather than appeal to that general theory.

Lemma 1 (Completion Lemma). *Let F_1, F_2 be probability measures on $\mathbb{R}$ and let μ be a finite measure on $\mathbb{R}^2$ with total mass $p \in (0, 1)$ and marginals μ_1, μ_2 satisfying $\mu_1 \leq F_1$ and $\mu_2 \leq F_2$. Then*

$$F' := \mu + \frac{(F_1 - \mu_1) \otimes (F_2 - \mu_2)}{1 - p}$$

is a probability measure on $\mathbb{R}^2$ with marginals exactly F_1 and F_2 , satisfying $\mu \leq F'$. Consequently $\pi' := d\mu/dF'$ is a valid selection function ($0 \leq \pi' \leq 1$) whose stayer measure is exactly μ and whose retention rate is p .

Proof. $(F_1 - \mu_1)$ and $(F_2 - \mu_2)$ are nonnegative measures (by the domination hypotheses) each of mass $1 - p$, so their product has mass $(1 - p)^2$ and the second term has mass $1 - p$; F' has mass 1. Its first marginal is $\mu_1 + (F_1 - \mu_1) \cdot \frac{(F_2 - \mu_2)(\mathbb{R})}{1 - p} = \mu_1 + (F_1 - \mu_1) = F_1$, and symmetrically for the second. Domination $\mu \leq F'$ holds because the added term is nonnegative. The Radon–Nikodym derivative $\pi' = d\mu/dF'$ exists and lies in $[0, 1]$ by domination; by construction $\pi' dF' = d\mu$, so the stayer measure is μ and $\int \pi' dF' = p$. $\square$

The completion has a design interpretation: the non-stayers are assigned the *independent* coupling of the leftover marginals. Nothing in the data restricts the dependence structure of those never observed at wave 2, and the lemma exploits exactly that slack.

Theorem 1 (Sharp identified set for the conditioning map). *Under refreshment validity, data*

compatibility $pG_1 \leq F_1$ (the stayers' first-wave sub-distribution fits under the full cohort's, which holds automatically when the first wave is observed for the whole cohort), and $c \in \mathcal{M}$, the identified set is exactly

$$\mathcal{C}_{ID} = \{c' \in \mathcal{M} : p \cdot Q_{c'} \leq F_2\},$$

and for every $c' \in \mathcal{C}_{ID}$ the rationalizing pair (F', π') can be taken to be the completion of $\mu_{c'} := (\text{id} \times c'^{-1})_{\#}(pG)$ given by Lemma 1. Maps are identified as elements of $\mathcal{M}$ up to Q -null sets—that is, on the support of the observed stayer distribution, almost surely—and not on intervals the data never visit.

Proof. Necessity. Let (F', π', c') rationalize $\mathcal{O}$. Since c' is a continuous strict increase, it is invertible on its range, and the stayers' latent joint measure is pinned by the data and c' :

$$\mu' = (\text{id} \times c'^{-1})_{\#}(pG) = \mu_{c'},$$

because the observed stayer measure of (Y_1, Y_2) is pG and $Y_2^* = c'^{-1}(Y_2)$ almost surely among stayers. Selection probabilities cannot exceed one: $\mu' = \pi' dF' \leq dF'$ setwise. Taking second marginals, $pQ_{c'} = \mu'_2 \leq F'_2 = F_2$, where the last equality is rationalization of the refreshment marginal. This is the stated constraint. (The first-marginal condition $pG_1 \leq F_1$ holds automatically in any data set generated by a panel, since the stayers' wave-1 sub-measure is part of the cohort's wave-1 measure.)

Sufficiency. Let $c' \in \mathcal{M}$ satisfy $pQ_{c'} \leq F_2$. Set $\mu = \mu_{c'}$; its mass is p , its first marginal is $pG_1 \leq F_1$ (automatic, as above), its second is $pQ_{c'} \leq F_2$ (the hypothesis). Lemma 1 delivers (F', π') with marginals (F_1, F_2) , stayer measure $\mu_{c'}$, and retention p . It remains to check that (F', π', c') reproduces $\mathcal{O}$: F_1, F_2 are the marginals of F' by construction; the stayer joint of $(Y_1, c'(Y_2^*))$ is $(\text{id} \times c')_{\#}\mu_{c'}/p = G$; and the retention rate is p . $\square$

Three remarks before the corollaries. First, the theorem says the stayers' *joint* distribution G carries no identifying information about c beyond its outcome marginal Q : for this question, the association between baseline and later reports is summarized by one distribution. Second,

the constraint is a *likelihood-ratio* condition, not a condition on distribution functions: where densities exist it reads $p q_{c'}(y) \leq f_2(y)$ for almost every y —domination must hold at every point of the support, which is more demanding than stochastic ordering. Third, p is observed, which distinguishes our problem from missing-data settings where the selection rate itself must be bounded.

3.1 Corollaries: what attrition does to identification

Corollary 1 (No attrition implies nonparametric point identification). *If $p = 1$ then $\mathcal{C}_{ID} = \{c\}$: the conditioning map is point-identified, with*

$$c = \text{the increasing rearrangement transporting } F_2 \text{ to } Q, \quad c(y) = Q^{-1}(F_2(y))$$

(writing CDFs for the measures), with no restriction on the dependence between attrition and outcomes—vacuously, since there is none.

Proof. With $p = 1$, the constraint $Q_{c'} \leq F_2$ holds between two probability measures, and setwise domination between measures of equal total mass forces equality: $Q_{c'} = F_2$. Thus c'^{-1} transports Q to F_2 ; for atomless distributions the increasing transport is unique, so c' is the stated rearrangement, which is also the true c (which satisfies the same equation). $\square$

Corollary 1 is the no-attrition benchmark against which existing identification results for “trained” versus fresh respondents (Das, Toepoel, and Soest 2011) can be compared: *if nobody left*, matching quantiles of the stayer and refreshment distributions recovers the conditioning map exactly, nonparametrically, item by item. Under the maintained deterministic monotone-map model and refreshment validity, attrition is what weakens the identifying restriction.

Corollary 2 (The identification funnel). *Let densities exist, and let $\bar{\pi}(y) := E[\pi(Y_1, Y_2^*) \mid Y_2^* = y]$ be the outcome-conditional selection function of the true structure, so that the stayers’ latent-outcome density satisfies $p q_c(y) = \bar{\pi}(y) f_2(y)$, where q_c denotes the density of*

$Q_c = (c^{-1})_{\#}Q$. Then $c' \in \mathcal{C}_{ID}$ if and only if

$$\bar{\pi}(g(y)) f_2(g(y)) g'(y) \leq f_2(y) \quad \text{for a.e. } y, \quad g := c^{-1} \circ c',$$

the Jacobian g' entering because the constraint compares measures, not ordinates (the density form requires g to be absolutely continuous with $g' > 0$ a.e., which we assume here in addition to membership in $\mathcal{M}$; the measure form of Theorem 1 does not). For a location perturbation $g(y) = y + \varepsilon$ the condition reads $\Phi(\varepsilon) \leq 1$, where

$$\Phi(\varepsilon) := \operatorname{ess\,sup}_y \frac{\bar{\pi}(y + \varepsilon) f_2(y + \varepsilon)}{f_2(y)}.$$

In particular: (i) $\varepsilon = 0$ always satisfies it, since $\Phi(0) = \operatorname{ess\,sup} \bar{\pi} \leq 1$; (ii) if $\bar{\pi} \geq \pi_{\min} > 0$ a.e., every feasible ε satisfies $R(\varepsilon) \leq 1/\pi_{\min}$, where R is the shift-ratio function of Corollary 3; hence if $\log f_2(y)/|y| \rightarrow -\infty$ as $y \rightarrow -\infty$ (as $y \rightarrow +\infty$), no $\varepsilon > 0$ (no $\varepsilon < 0$) is feasible; (iii) if $\operatorname{ess\,sup} \bar{\pi} \leq 1 - \kappa$ for some $\kappa \in (0, 1)$ and $\log f_2$ is L -Lipschitz, every ε with $|\varepsilon| \leq L^{-1} \log\{1/(1 - \kappa)\}$ is feasible. The condition $\Phi(\varepsilon) \leq 1$ is global in y : it can fail in the centre of the distribution although $\bar{\pi}$ decays in both tails (Example below).

Proof. The candidate latent distribution is $Q_{c'} = (c'^{-1})_{\#}Q = (c'^{-1} \circ c)_{\#}Q_c = (g^{-1})_{\#}Q_c$. Its density follows from the change-of-variables identity $\int_A q_{c'} dy = Q_c(g(A)) = \int_A q_c(g(y)) g'(y) dy$ for every Borel A , so $q_{c'}(y) = q_c(g(y)) g'(y)$ a.e. Substituting $p q_c = \bar{\pi} f_2$, the sharp constraint $p q_{c'}(y) \leq f_2(y)$ a.e. (Theorem 1 in density form) is exactly the display, and with $g(y) = y + \varepsilon$ it is $\Phi(\varepsilon) \leq 1$. (i) is immediate. (ii) $\Phi(\varepsilon) \geq \pi_{\min} R(\varepsilon)$, and $R(\varepsilon) = \infty$ for every $\varepsilon > 0$ under the lower-tail condition (and for every $\varepsilon < 0$ under the upper-tail condition) by the telescoping argument in the proof of Corollary 3, which uses only f_2 . (iii) $\log f_2(y + \varepsilon) - \log f_2(y) \leq L|\varepsilon|$ for every y , so $\Phi(\varepsilon) \leq (1 - \kappa) e^{L|\varepsilon|} \leq 1$. $\square$

Corollary 2 makes precise how attrition and conditioning are confounded: a perturbation of the map is admissible exactly when the selection-weighted latent density, moved by the perturbation, still fits under the refreshment density at every outcome value. Parts (ii) and (iii) give a necessary and a sufficient condition; neither is a tail criterion alone.

Example (central binding). Let $F_2 = N(0, 1)$, $Q = N(0, \sigma^2)$ with $\sigma \in (0, 1)$, $p = \sigma$, and let the true map be the identity. The implied selection function $\bar{\pi}(y) = p q(y)/f_2(y) = \exp\{-(\sigma^{-2} - 1)y^2/2\}$ lies in $(0, 1]$ and decays in both tails, and the stayers' latent density has strictly lighter tails than f_2 on both sides. Yet for every $\varepsilon \neq 0$,

$$\Phi(\varepsilon) = \sup_y \frac{p q(y + \varepsilon)}{f_2(y)} = \exp\left\{\frac{\varepsilon^2}{2(1 - \sigma^2)}\right\} > 1,$$

the supremum being attained at $y = -\varepsilon/(1 - \sigma^2)$: the identified set is $\{0\}$ ($\sigma = p = .5$ gives a concrete structure). The constraint binds at the centre, where $\bar{\pi} = 1$, although there is slack in both tails. The example also shows that non-identification requires an actual feasible $\varepsilon \neq 0$: here $\inf_{\varepsilon \neq 0} \Phi(\varepsilon) = 1$, but the infimum is not attained away from zero.

Example (logistic selection). Let f_2 be Gaussian with variance σ^2 and let the true selection be logistic in the outcome, $\bar{\pi}(y) = \Lambda(\alpha + \beta y)$ with $\beta > 0$ (better-off respondents stay). For a location perturbation $c'(y) = c(y) + \Delta$ the constraint reads $\Lambda(\alpha + \beta(y + \Delta)) f_2(y + \Delta) \leq f_2(y)$ for all y . No $\Delta < 0$ is feasible (the constraint fails in the upper tail, where $\bar{\pi} \rightarrow 1$), and in the lower tail the log-ratio behaves as $y(\beta - \Delta/\sigma^2)$, so no $\Delta > \beta\sigma^2$ is feasible either. Because f_2 is unimodal, the set is an interval (Theorem 3), $[0, \Delta^*]$ with

$$\Delta^* \leq \beta\sigma^2,$$

and the endpoint depends on the retention level through α : the interior maximum of the constraint ratio (attained where $1 - \Lambda = \Delta/\beta\sigma^2$) pushes Δ^* towards 0 as retention rises ($\alpha \rightarrow \infty$, where the set must collapse, since $p \rightarrow 1$ recovers Corollary 1) and towards the envelope $\beta\sigma^2$ as retention falls. The envelope is attained once retention is low enough that the lower-tail limit of the constraint ratio, $\exp\{\alpha + \beta\Delta - \Delta^2/(2\sigma^2)\}$ evaluated at $\Delta = \beta\sigma^2$, is at most one—that is, $\alpha \leq -\beta^2\sigma^2/2$ —and the interior supremum is also at most one (with $\sigma^2 = \beta = 1$ and $\alpha = -1$, retention .30, the supremum at $\Delta = 1$ is $e^{-1/2}$, so $\Delta^* = \beta\sigma^2$ exactly). In this model, selection strength (β) and outcome dispersion (σ^2) multiply to produce the ceiling on the ambiguity: with mild outcome selection ($\beta\sigma \approx 0.1$ in standardized units) at most a tenth of a standard deviation of conditioning can be confused

with selection, whatever the retention rate. This is a continuous-outcome counterpart of the worst-case bounds of Das, Toepoel, and Soest (2011), which rest on the bounded-outcome logic of Manski (1989, 345–46, eqs. (5)–(6)): their bounds have width equal to the attrition rate because they let the attriters’ unobserved answers range over the whole outcome space, whereas Corollary 2 requires the attriters’ implied density to fit under f_2 pointwise. The example is stated for location shifts $c(y) = y + h$; for a general map c the perturbation $c' = c + \Delta$ does not reduce to $g(y) = y + \Delta$, and the constraint must be evaluated on $g = c^{-1} \circ c'$ directly.

Corollary 3 (Attrition independent of the outcome). *Let the conditioning map be a location shift, $c(y) = y + h$, and suppose the true selection does not depend on the latent outcome, $\bar{\pi}(y) \equiv p$, so that the stayers’ latent distribution equals the population’s, $Q_c = F_2$. Define the shift-ratio function of the refreshment density,*

$$R(\varepsilon) := \operatorname{ess\,sup}_y \frac{f_2(y + \varepsilon)}{f_2(y)} \in [1, \infty], \quad R(0) = 1.$$

Then the identified set of Theorem 1 for the shift is

$$\mathcal{H}_{ID} = h + \{ \varepsilon : R(\varepsilon) \leq 1/p \}.$$

In particular:

- (i) *if $\log f_2(y)/|y| \rightarrow -\infty$ as $y \rightarrow -\infty$ and as $y \rightarrow +\infty$ (tails thinner than exponential on both sides, as for the Gaussian), then $R(\varepsilon) = \infty$ for every $\varepsilon \neq 0$ and $\mathcal{H}_{ID} = \{h\}$: the shift is point identified at every retention rate $p < 1$, although attrition is present and its mechanism is unknown to the analyst; if the condition holds in the lower (upper) tail only, no $\varepsilon > 0$ ($\varepsilon < 0$) belongs to the set;*
- (ii) *if $\log f_2(y)/|y| \rightarrow -\lambda$ as $y \rightarrow \pm\infty$ for some $\lambda > 0$, then $R(\varepsilon) \geq e^{\lambda|\varepsilon|}$ and $\mathcal{H}_{ID} \subseteq [h - \lambda^{-1} \log(1/p), h + \lambda^{-1} \log(1/p)]$; if in addition $\log f_2$ is λ -Lipschitz—as for the Laplace density $f_2(y) \propto e^{-\lambda|y|}$ and, with $\lambda = 1$, the standard logistic density—then $R(\varepsilon) = e^{\lambda|\varepsilon|}$ and the inclusion is an equality, an interval of width $2\lambda^{-1} \log(1/p)$ that vanishes as $p \rightarrow 1$ and grows without bound as $p \rightarrow 0$;*

(iii) if $\log f_2$ is L -Lipschitz, $\mathcal{H}_{ID} \supseteq [h - L^{-1} \log(1/p), h + L^{-1} \log(1/p)]$, and if f_2 is unimodal, $\mathcal{H}_{ID}$ is an interval (Theorem 3). For the standard Cauchy density, $R(\varepsilon) = \left(\frac{|\varepsilon|}{2} + \sqrt{1 + \varepsilon^2/4}\right)^2$ and $\mathcal{H}_{ID} = [h - (1-p)/\sqrt{p}, h + (1-p)/\sqrt{p}]$. Tail behaviour alone does not determine $\mathcal{H}_{ID}$: a smooth, strictly positive density with Cauchy tails, or with logistic tails, can have a disconnected identified set.

Proof. With $Q_c = F_2$ the reported density is $q(y) = f_2(y - h)$. A candidate $c'(y) = y + h'$ implies the latent density $q_{c'}(y) = q(y + h') = f_2(y + \varepsilon)$ with $\varepsilon := h' - h$, and the constraint of Theorem 1, $p q_{c'} \leq f_2$ a.e., reads $p f_2(y + \varepsilon) \leq f_2(y)$ for a.e. y , i.e. $R(\varepsilon) \leq 1/p$; sharpness is Theorem 1's. Parts (i) and (ii) rest on a telescoping bound. If $R(\varepsilon) = M < \infty$ for some $\varepsilon > 0$, then $f_2(y) \geq f_2(y + \varepsilon)/M$ for a.e. y ; the set of y_0 for which some $y_0 - n\varepsilon$, $n \geq 1$, falls in the exceptional null set is null, so for a.e. y_0 iteration gives $\log f_2(y_0 - n\varepsilon) \geq \log f_2(y_0) - n \log M$ for every n , and hence $\liminf_{n \rightarrow \infty} \log f_2(y_0 - n\varepsilon)/(n\varepsilon) \geq -\varepsilon^{-1} \log M$. (i) Under the lower-tail condition the left side is $-\infty$, a contradiction; so $R(\varepsilon) = \infty$ for every $\varepsilon > 0$, and the case $\varepsilon < 0$ uses the upper tail in the same way. (ii) Under the tail condition the left side equals $-\lambda$, so $\log M \geq \lambda\varepsilon$; symmetrically for $\varepsilon < 0$. If $\log f_2$ is λ -Lipschitz, $\log f_2(y + \varepsilon) - \log f_2(y) \leq \lambda|\varepsilon|$ for every y , so $R(\varepsilon) = e^{\lambda|\varepsilon|}$. The Laplace log-density $-\lambda|y|$ is λ -Lipschitz with tail rate λ ; the logistic log-density has derivative $-\tanh(y/2) \in (-1, 1)$ and tail rate one. (iii) The Lipschitz bound gives $R(\varepsilon) \leq e^{L|\varepsilon|}$, and the interval statement is Theorem 3 with $q(\cdot) = f_2(\cdot - h)$. For $f_2(y) \propto (1 + y^2)^{-1}$, the ratio $r(y) = (1 + y^2)/(1 + (y + \varepsilon)^2)$ is continuous and tends to one at both tails; $r'(y) = 0$ reduces to $y^2 + \varepsilon y - 1 = 0$, and at $y^* = -\frac{1}{2}(\varepsilon + \sqrt{\varepsilon^2 + 4})$ one finds $r(y^*) = \left(\frac{\varepsilon}{2} + \sqrt{1 + \varepsilon^2/4}\right)^2 = e^{2 \operatorname{asinh}(\varepsilon/2)}$ for $\varepsilon \geq 0$, so $R(\varepsilon) \leq 1/p$ iff $|\varepsilon| \leq 2 \sinh\{\frac{1}{2} \log(1/p)\} = (1 - p)/\sqrt{p}$. For the last sentence, let $f(y) \propto (1 + y^2)^{-1} \exp\{2e^{-y^2} \cos(20\pi y)\}$ and $p = 1/2$. The modulation has period 0.1, so for $\varepsilon = 0.1$ the log-ratio is at most $0.1 + 0.2\sqrt{2/e} < \log 2$ and 0.1 is feasible; but at $y = 0.05$ the log-ratio for $\varepsilon = 0.05$ is $\log f(0.1) - \log f(0.05) \approx 3.97 > \log 2$, so 0.05 is not. Replacing $(1 + y^2)^{-1}$ by the logistic density gives the same conclusion with exponential tails, and there $R(1) \neq e$ although $R(\varepsilon) \geq e^{|\varepsilon|}$ (Appendix, deterministic checks). $\square$

Under outcome-independent attrition, then, the identified set is a sublevel set of the refreshment density's shift-ratio function. What identifies the shift is not the benignity of the

attrition—which the analyst cannot see—but a property of the refreshment density, which she can estimate. Three consequences follow. First, the width of the set is not a measure of selection: for the Laplace, logistic and Cauchy densities, and more generally whenever $\log f_2$ is Lipschitz, there is a nondegenerate set even when attrition is completely ignorable, while with tails thinner than exponential there is none even at low retention; any plug-in width must be read with this in mind (Remark 2). Second, under outcome-dependent attrition, whether the set is degenerate depends on the whole function Φ of Corollary 2 and not on tails alone: nondegeneracy means that $\Phi(\varepsilon) \leq 1$ for some $\varepsilon \neq 0$, which is the hypothesis of Theorem 4(b). Third, selection cannot turn a Gaussian population into a Gaussian of the same variance with a different mean: the ratio $p \exp(\delta y - \delta^2/2)$ of such a stayer density to the population density is unbounded in the tail towards which the mean moves, so no admissible selection process realizes it, and under outcome-independent attrition the Gaussian is a case of point identification.

Remark 1. On what a funnel measures. Corollary 2 gives the funnel’s shape under a specified selection; Corollary 3 gives it under no selection on the outcome. Reading the two together: a perturbation is admissible where the *stayers’* latent density, moved by it, still fits under f_2 , and that room comes from two sources that the data do not distinguish—selection that keeps the stayers’ latent density below f_2 wherever the moved density must fit, and a refreshment density whose shape can absorb a shift. Only the second is visible from the refreshment sample alone.

Remark 2. Point identification from the far tails, and what a trimmed support does to it. Part (i) of Corollary 3 rests on the extreme tails: for a Gaussian f_2 with variance σ^2 and a shift $\varepsilon > 0$, the constraint $p f_2(y + \varepsilon) \leq f_2(y)$ fails only where $y \leq -\{\sigma^2 \log(1/p)/\varepsilon + \varepsilon/2\}$, that is, at a distance of order $\sigma^2 \log(1/p)/\varepsilon$ into the tail. Small shifts are excluded only far out, where no finite sample has data. If the constraint is imposed only on $|y| \leq Y$ —as any plug-in implementation must, for example by trimming the support at a fraction of the mode as in Simulation 1—the effective set for a shift under outcome-independent attrition is not $\{h\}$ but the interval $|\varepsilon| \lesssim \sigma^2 \log(1/p)/Y$ (to first order in ε), and a slack κ in the criterion, $p q_{c'} \leq (1 + \kappa) f_2$, widens it to $\sigma^2 \log\{(1 + \kappa)/p\}/Y$. With $Y = 2.15\sigma$ (the 10%-of-mode

trim), $p = .8$, and $\kappa = .10$, this gives a half-width of $.15\sigma$; the exact population version of the same trimmed and relaxed criterion has width $.31\sigma$, against mean plug-in widths of .25–.26 in Simulation 1 at $\beta = 0$, $p = .8$, where the identified set is a point. The plug-in width is thus, to a good approximation, a statement about the trim and the slack, not about selection, and a positive numerical width is not by itself an identification deficit. This is why Section 10 treats plug-in criteria only as exploratory checks.

3.2 Discrete outcomes

The class $\mathcal{M}$ is rich when the outcome is continuous and nearly empty when it is discrete, and the difference decides how the theorem is used on survey items. Let the outcome take values in a finite ordered set $\mathcal{Y} = \{y_1 < \dots < y_m\}$ and let conditioning be a *weakly* increasing map $c : \mathcal{Y} \rightarrow \mathcal{Y}$, so that maps merging adjacent categories are allowed. Because a map may merge, the stayers' latent sub-distribution is no longer pinned by their reports, and a structure must specify it; the identified set is defined as before, as the set of maps for which some structure rationalizes the observables.

Corollary 4 (Discrete outcomes: mass domination). *Under refreshment validity, a weakly increasing c' belongs to the identified set if and only if*

$$pQ(\{a\}) \leq F_2(c'^{-1}(\{a\})) \quad \text{for every } a \in \mathcal{Y} : \quad (1)$$

the retention-scaled mass the stayers report in each category must fit under the population mass of the latent categories the map sends there. Consequently: (i) the image of c' must contain the set $\mathcal{Y}_Q$ of categories the stayers report, $\mathcal{Y}_Q \subseteq c'(\mathcal{Y})$; categories in the image that the stayers do not report impose no constraint; (ii) when the stayers report every category, c' is onto and hence the identity, and the identified set is $\{\text{id}\}$ if $pQ(\{a\}) \leq F_2(\{a\})$ for every a and empty otherwise; (iii) for a binary item the same holds whenever both values are reported. When a category goes unreported, the identity can remain feasible and non-identity maps can be feasible, even uniquely: with F_2 uniform on $\{0, 1, 2\}$, $p = 1$, and $Q = (2/3, 0, 1/3)$, the only feasible map sends $(0, 1, 2)$ to $(0, 0, 2)$. In case (ii) the deterministic model has no identifying content beyond a specification test: the inequality $\max_a pQ(\{a\})/F_2(\{a\}) \leq 1$ is

a sharp implication of “no conditioning” that holds under every attrition process. Its violation is evidence against the maintained model—refreshment validity, comparable coding of the item in the panel and the refreshment sample, and deterministic monotone measurement—and, if the first two are credible, evidence of conditioning that must be modelled as a stochastic change of individual answers (the model of Das, Toepoel, and Soest (2011) for binary items) rather than as a deterministic map.

Proof. Necessity. A rationalizing structure has a stayers’ latent sub-measure ν on $\mathcal{Y}$ (the second marginal of the stayers’ latent joint) with $\nu \leq F_2$, because selection probabilities are at most one, and with $c'_\# \nu = pQ$; hence $pQ(\{a\}) = \nu(c'^{-1}(\{a\})) \leq F_2(c'^{-1}(\{a\}))$. *Sufficiency.* On each block $c'^{-1}(\{a\})$ let ν be F_2 restricted to the block and scaled by $pQ(\{a\})/F_2(c'^{-1}(\{a\})) \leq 1$ (zero if $pQ(\{a\}) = 0$); then $\nu \leq F_2$ and $c'_\# \nu = pQ$. Couple ν with the entry report by giving the stayers whose latent category b lies in the block $c'^{-1}(\{a\})$ the observed conditional law of Y_1 given the report a : the joint stayer measure $\mu(dy_1 \times \{b\}) := G(dy_1 \mid Y_2 = a) \nu(\{b\})$ has first marginal $\sum_a G(dy_1 \mid Y_2 = a) pQ(\{a\}) = pG_1 \leq F_1$ and second marginal $\nu \leq F_2$, and Lemma 1, which holds on any measurable space, completes the structure when $p < 1$; when $p = 1$ the constraints force $\nu = F_2$, and μ is itself the structure. (i): a category outside the image of c' has empty preimage, so Equation 1 forces $pQ(\{a\}) = 0$ there; every reported category therefore lies in the image, while an unreported category in the image satisfies Equation 1 trivially. (ii): if every category is reported, the image is $\mathcal{Y}$, and a weakly increasing map of a finite chain onto itself is strictly increasing, hence the identity. (iii) is (ii) with $m = 2$. In the example, Equation 1 at $a = 0$ requires the preimage of 0 to carry mass at least 2/3, hence to be $\{0, 1\}$, and at $a = 2$ requires 2 in the preimage of 2. The “every attrition process” clause is the necessity direction, which uses only $\nu \leq F_2$. $\square$

The corollary settles the scope of the deterministic model on the two kinds of survey item. For continuous items, and for ordinal items with enough categories to be treated as continuous, the class of maps is rich and the funnel is the object of interest. For items with a handful of categories, all of which the stayers use—most attitude and behaviour items in panels of the size considered here—the only admissible deterministic map is the identity, so what

Theorem 1 delivers is a *test*: the retention-scaled stayer mass in every category must fit under the refreshment mass. Section 10 applies exactly this criterion. The response-style conditioning that the companion empirical paper documents—extreme categories abandoned for the midpoint by some respondents—is, in these terms, a stochastic partial merging of categories that no deterministic map represents (a map sending 5 to 4 would have to do so for everyone, and the stayers still report 5), and for which the appropriate identification analysis is the bounds analysis of Das, Toepoel, and Soest (2011) extended from binary to ordered categories, which we do not pursue. The check of Section 12 enumerates every weakly increasing map on four categories and confirms Equation 1 against a linear-programming feasibility test.

4 Multiwave Panels: What the Longitudinal Structure Adds

Theorem 1 is a two-wave statement. A panel observes a cohort at many waves, with attrition unfolding between them, with a battery of items asked once at entry, and with response patterns that may be interrupted; a natural conjecture is that this longitudinal structure—the joint distribution of reports across waves, the timing of dropout, the negative controls—narrows the identified set of the conditioning map at any given wave. It does not. This section shows that, within one cohort and without restrictions that link waves, the identified set of the map at a wave with a refreshment sample is exactly the two-wave set of Theorem 1 computed from that wave’s respondents and its refreshment distribution, and that the maps at waves without a refreshment sample are unidentified. The result supplies the multiwave construction that the necessity part of Theorem 4 requires, and it says precisely what the constructive designs of Section 7 must assume: a restriction linking the selection at one wave to something observed at another.

4.1 Setting

A cohort enters at wave e and is followed for s waves; index waves by tenure $j = 1, \dots, s$. The latent path is $Y^* = (Y_1^*, \dots, Y_s^*) \sim F$ on $\mathbb{R}^s$; the entry coordinate may be vector-valued, carrying any battery of items asked once at entry, and nothing below changes if it is. Conditioning maps are tenure-specific, $c_1 = \text{id}$ and $c_j \in \mathcal{M}$ for $j \geq 2$, and reports are $Y_j = c_j(Y_j^*)$; the case in which the map depends on the number of completed interviews rather than on tenure is treated in Remark 3. A *response pattern* is $r \in \{0, 1\}^s$ with $r_1 = 1$; monotone attrition is the case $r = (1, \dots, 1, 0, \dots, 0)$, but interrupted patterns are allowed. Selection is a Markov kernel $\pi(r \mid y^*) = \Pr(R = r \mid Y^* = y^*)$, unrestricted: the probability of any pattern may depend on the whole latent path, including coordinates that are never reported. Write $p_r := \Pr(R = r)$ and $p_j := \sum_{r: r_j=1} p_r$ for the response rate at tenure j .

The observables are, for each pattern r , the sub-measure λ_r on $\mathbb{R}^{|r|}$ of the reported coordinates $(Y_j)_{j: r_j=1}$ among respondents with that pattern (of mass p_r); and, at every tenure j in a set J^* at which a fresh cohort is drawn, the population latent marginal F_j (refreshment validity and cohort comparability, (A1) below). The entry marginal $F_1 = \sum_r \text{proj}_1 \lambda_r$ is observed automatically. Let $Q_j := p_j^{-1} \sum_{r: r_j=1} \text{proj}_j \lambda_r$ be the reported-outcome distribution of all respondents at tenure j , whatever their pattern, and $Q_{j,c'} := (c'^{-1})_{\#} Q_j$ its pullback through a candidate map. A structure is $\theta = (F, \pi, c_2, \dots, c_s)$; it rationalizes the observables if it reproduces every λ_r and every F_j , $j \in J^*$.

Lemma 2 (Pattern completion). *Let $\{\tilde{\lambda}_r\}$ be finite measures, $\tilde{\lambda}_r$ on the coordinates $\{j : r_j = 1\}$, indexed by patterns with $r_1 = 1$ and with masses summing to one. Let $J^* \subseteq \{1, \dots, s\}$ and, for $j \in J^*$, let F_j be a probability measure on $\mathbb{R}$ with $\sum_{r: r_j=1} \text{proj}_j \tilde{\lambda}_r \leq F_j$. Then there exist a probability measure F' on $\mathbb{R}^s$ and a Markov kernel $\pi'(r \mid y^*)$ such that (i) $\text{proj}_j F' = F_j$ for every $j \in J^*$ and $\text{proj}_1 F' = \sum_r \text{proj}_1 \tilde{\lambda}_r$; and (ii) for every pattern r , the sub-measure of the coordinates $\{j : r_j = 1\}$ among those with $R = r$ under (F', π') is exactly $\tilde{\lambda}_r$.*

Proof. For $j \notin J^*$ fix any probability measure κ_j on $\mathbb{R}$. For $j \in J^*$ let p_j be the total mass of $\sum_{r: r_j=1} \tilde{\lambda}_r$; if $p_j < 1$ set $L_j := F_j - \sum_{r: r_j=1} \text{proj}_j \tilde{\lambda}_r$, a nonnegative measure of mass $1 - p_j$ by hypothesis, and $\kappa_j := L_j / (1 - p_j)$; if $p_j = 1$ nobody is missing at j and κ_j is not needed.

For each pattern r define the measure on $\mathbb{R}^s$

$$\nu_r := \tilde{\lambda}_r \otimes \bigotimes_{j:r_j=0} \kappa_j,$$

which gives the reported coordinates the law $\tilde{\lambda}_r$ and the unreported ones independent draws from the κ_j . Put $F' := \sum_r \nu_r$, a probability measure because the masses sum to one, and $\pi'(r \mid y^*) := (d\nu_r/dF')(y^*)$, which exists and lies in $[0, 1]$ because $\nu_r \leq F'$, and which sums over r to $dF'/dF' = 1$. The pattern- r sub-measure is then $\pi'(r \mid \cdot) F' = \nu_r$, whose projection onto the reported coordinates is $\tilde{\lambda}_r$: this is (ii). For (i), $\text{proj}_j F' = \sum_{r:r_j=1} \text{proj}_j \tilde{\lambda}_r + \sum_{r:r_j=0} p_r \kappa_j = (F_j - L_j) + (1 - p_j) L_j / (1 - p_j) = F_j$ for $j \in J^*$, since $\sum_{r:r_j=0} p_r = 1 - p_j$; at $j = 1$ every pattern reports, so $\text{proj}_1 F' = \sum_r \text{proj}_1 \tilde{\lambda}_r$. $\square$

Under monotone attrition the kernel takes the sequential form used in Section 7: writing $N_j := \sum_{r:r_j=1} \nu_r$ for the sub-measure of those still responding at tenure j , one has $N_1 = F' \geq N_2 \geq \dots \geq N_s$, and $\pi'_j := dN_{j+1}/dN_j \in [0, 1]$ is the probability of continuing past j given the latent path; the survivors through j then have latent sub-measure $\prod_{u < j} \pi'_u F' = N_j$. The completion is the multiwave form of Lemma 1: everyone's unreported coordinates are assigned independently of their reported ones, with the leftover distribution at each refreshment wave, and nothing in the data objects.

Theorem 2 (No additional identification from within-cohort histories under unrestricted cross-wave dependence). *Under refreshment validity and cohort comparability at the tenures in J^* , the identified set for the family of maps $(c_j)_{j \in J^*}$ is the product*

$$\prod_{j \in J^*} \{c' \in \mathcal{M} : p_j Q_{j,c'} \leq F_j\},$$

each factor the two-wave set of Theorem 1 computed from the tenure- j respondents' reported distribution, the response rate at j , and the refreshment distribution at j . The maps at tenures outside J^ are unidentified, every element of $\mathcal{M}$ being compatible with the data; and the joint distribution of reports across waves, the response patterns, and any battery of entry-wave items impose no further restriction. In particular, for a panel with one refreshment sample*

at tenure s , the identified set of c_s is exactly $\{c' : p_s Q_{s,c'} \leq F_s\}$, whatever the panel's length, its attrition history, and its entry battery.

Proof. Necessity. Let (F', π', c') rationalize the observables and fix $j \in J^*$. The latent sub-measure of the tenure- j respondents is $\sum_{r:r_j=1} \pi'(r \mid \cdot) F' \leq F'$, because the kernel's values are nonnegative and sum to one over r ; its j -th marginal is therefore dominated by $\text{proj}_j F' = F_j$. By rationalization that marginal is also the pullback of the observed reported distribution, $(c_j'^{-1})_{\#}(p_j Q_j) = p_j Q_{j,c_j'}$. Hence $p_j Q_{j,c_j'} \leq F_j$.

Sufficiency. Take candidates c'_j , $j \in J^*$, each satisfying its constraint, arbitrary $c'_j \in \mathcal{M}$ at the other tenures, and $c'_1 = \text{id}$. Pull every observed pattern measure back to the latent scale coordinate by coordinate, $\tilde{\lambda}_r := (\times_{j:r_j=1} c_j'^{-1})_{\#} \lambda_r$. For $j \in J^*$, $\sum_{r:r_j=1} \text{proj}_j \tilde{\lambda}_r = (c_j'^{-1})_{\#}(p_j Q_j) = p_j Q_{j,c_j'} \leq F_j$, which is the hypothesis of Lemma 2. The lemma returns (F', π') ; pushing each ν_r 's reported coordinates forward through the c'_j recovers λ_r , and $\text{proj}_j F' = F_j$ on J^* . So (F', π', c') rationalizes the observables. The product structure follows because the constraints, and the construction, involve each $j \in J^*$ separately, and the unrestricted tenures never enter a constraint. $\square$

The theorem extends the first remark after Theorem 1—that, for identifying c , the stayers' joint distribution with their baseline reports is summarized by one distribution—to the entire panel: a cohort's reports over any number of waves, the pattern of its dropouts, and any battery of entry items are exhausted, for identifying the map at a wave with a refreshment sample, by that wave's reported distribution, its response rate, and its refreshment distribution. What the longitudinal structure buys is not identification of c but the possibility of *restrictions* that use it. The transport restriction (A2) of Section 6 links the target item's selection at t to the negative controls' selection at t ; the restrictions B2–B4 of Section 7 link the survivors' latent mean at t to something observed—the fresh cohort's future or the continuing cohort's past. Each of these is violated by the construction in the proof, which shifts the survivors' latent distribution at t while holding every entry-wave quantity and every earlier wave fixed; that is exactly why each of them identifies. The theorem also places the fixed-effects results of the companion identification paper: the additive cell-mean structure

there is a restriction across cohorts and waves of the kind the present theorem says is needed, and the stride theorem there says how much of the path such a restriction recovers.

Corollary 5 (Repeated tenures). *In the tenure model, where c_s is common to all cohorts, suppose tenure s is observed at two refreshment episodes—cohorts $e_1 < e_2$ compared with fresh cohorts at $t_r = e_r + s - 1$, $r = 1, 2$. Then the identified set of c_s is the intersection of the two episodes' sets, $\bigcap_{r=1,2} \{c' : p^{(r)} Q_{c'}^{(r)} \leq F^{(r)}\}$, with $p^{(r)}$, $Q^{(r)}$, $F^{(r)}$ the response rate, reported distribution, and refreshment distribution of episode r . It contains c_s , can be strictly smaller than either set, and is not in general a point. Under the additional hypothesis that the selection functional is the same at the two episodes, $\beta_1(s) = \beta_2(s)$ in the notation of Section 6.1, within the location family $c_s(y) = y + h$, and with the moments of the two episodes expressed on a common scale (raw units or a fixed reference standard deviation), the data carry the testable restriction $C_1(s) = C_2(s)$: if it holds, the identified set of the shift is unchanged by the hypothesis, and if it fails, the hypothesis is rejected. On episode-specific scales, or for general maps, the hypothesis can exclude maps that satisfy both episodes' domination constraints.*

Proof. The two cohorts are independent samples with separate latent processes and selection kernels, so Theorem 2 applies to each with the common candidate c'_s , which is admissible iff it is admissible at both episodes; c_s satisfies both constraints. That the intersection is not a point in general: under outcome-independent attrition at both episodes with Laplace refreshment tails of rate λ , Corollary 3 gives intervals of half-widths $\lambda^{-1} \log(1/p^{(r)})$ about h , whose intersection is the smaller interval. That it can be strictly smaller than either set: with Gaussian refreshment distributions and logistic selection increasing in the outcome at one episode and decreasing at the other, the example following Corollary 2 gives the one-sided sets $[h, h + a]$ and $[h - b, h]$ with $a, b > 0$, whose intersection is $\{h\}$. For the location clause, a candidate h' implies survivor latent means $\mathbb{E}[Y_t^{(r)} | S] - h'$ at both episodes; on a common scale, $\beta_1(s) = \beta_2(s)$ reads $C_1(s) = C_2(s)$ after h' cancels, a restriction on the data alone. If instead the episode- r moments are divided by different standard deviations σ_r , the restriction reads $(C_1(s) - h')/\sigma_1 = (C_2(s) - h')/\sigma_2$ in raw units, which pins h' when $\sigma_1 \neq \sigma_2$. For general maps the candidate does not cancel: with $F^{(1)} = Q^{(1)} = N(0, 1)$, $F^{(2)} = Q^{(2)} = N(1, 1)$,

retention .1 at both episodes and the true identity (selection differentials zero at both), the candidate $c'(y) = 2y$ has pullbacks $N(\mu_r/2, 1/4)$, whose largest density ratios against $F^{(r)}$ are $2\exp(\mu_r^2/6)$ with $\mu_r \in \{0, 1\}$, so it satisfies both constraints; but it implies survivor latent means $\mu_r/2$, hence selection differentials 0 and $-1/2$, and violates $\beta_1(s) = \beta_2(s)$. $\square$

The corollary is the formal content of the recommendation, made in Section 9, that panels engineer a repeated tenure across refreshment episodes: the repetition intersects two funnels and, in the location family, adds a direct test of stationary selection, but it does not by itself turn a funnel into a point. Point identification needs a restriction of the kind Theorem 4 and Section 7 supply.

Remark 3. Interrupted participation and dose-specific maps. If the map at a wave depends on the number k of interviews completed before it rather than on tenure—the dose model of the companion empirical paper—then respondents at tenure j with different patterns carry different maps $c_{j,k}$, and the constraint at $j \in J^*$ becomes $\sum_k (c'_{j,k})_{\#}(p_{j,k} Q_{j,k}) \leq F_j$, where $p_{j,k}$ and $Q_{j,k}$ are the response rate and reported distribution of the tenure- j respondents with dose k : a joint constraint on the family $\{c_{j,k}\}_k$ requiring the *sum* of the dose-specific pullbacks to fit under F_j . The identified set for the family is then not a product over k , but the proof is unchanged, since Lemma 2 is stated for arbitrary patterns. The companion identification paper’s assumption of uninterrupted participation (its M1) is the case in which every respondent at a tenure has the same dose, and the present theorem is the reason interrupted patterns cost nothing for the identification question addressed here: they are re-encoded, not used.

A deterministic check of the construction is reported in Section 12: on a four-wave cohort with a finite latent lattice, a binary entry-only negative control, all eight response patterns, and a selection kernel that depends on the never-reported coordinates, the completion of Lemma 2 is built explicitly for every candidate map at the refreshment wave; those satisfying the domination inequality reproduce every pattern sub-measure, the entry marginal, and the refreshment marginal within the script’s tolerance of 10^{-10} while moving the survivors’ latent mean by the candidate’s perturbation, and those violating it have a negative leftover.

5 Sharp Bounds for Location-Shift Conditioning

Before restricting the map, it is worth recording what Theorem 1 says about the survivors' mean conditioning effect when the map is unrestricted, because the answer is a known object and it is the benchmark against which the shape restriction should be judged.

Corollary 6 (Mean bounds for an unrestricted map). *Under the conditions of Theorem 1, suppose that $p < 1$, that F_2 and Q are atomless with support $\mathbb{R}$, and that both have finite means. Let $m^S := \mathbb{E}[Y_2^* \mid S = 1]$ be the survivors' latent mean and $h^S := \mathbb{E}[Y_2 - Y_2^* \mid S = 1]$ their mean conditioning effect. Write $q_u := F_2^{-1}(u)$ and*

$$m_L(p) := \mathbb{E}_{F_2}[Y \mid Y \leq q_p], \quad m_U(p) := \mathbb{E}_{F_2}[Y \mid Y \geq q_{1-p}]$$

for the means of the lower and upper p -fractions of the refreshment distribution. Then the identified set for m^S is the open interval $(m_L(p), m_U(p))$ and that for h^S is $(\mathbb{E}[Y_2 \mid S] - m_U(p), \mathbb{E}[Y_2 \mid S] - m_L(p))$: every interior point is attained by a map in $\mathcal{M}$, and the endpoints, which are the sharp infimum and supremum, are not. The closures are the trimmed-means intervals. The width $w(p) := m_U(p) - m_L(p)$ depends on the refreshment distribution and the retention rate only. The same argument bounds the survivors' latent distribution function pointwise: $\max\{0, (F_2(y) - 1 + p)/p\} \leq \Pr(Y_2^ \leq y \mid S = 1) \leq \min\{1, F_2(y)/p\}$.*

Proof. By Theorem 1 the survivors' latent distribution $\nu := Q_{c'}$ ranges over the probability measures with $p\nu \leq F_2$ that are pullbacks of Q through some $c' \in \mathcal{M}$. Any ν with $p\nu \leq F_2$ has $\int y d\nu \leq m_U(p)$: the density $d\nu/dF_2$ is bounded by $1/p$ and integrates to one, and among such densities the mean is maximized by the one equal to $1/p$ on $[q_{1-p}, \infty)$ and zero below, since moving mass from below q_{1-p} to above it, at density at most $1/p$, raises the mean; because F_2 is atomless this maximizer $\nu^* := p^{-1}F_2|_{[q_{1-p}, \infty)}$ is unique, and symmetrically $\int y d\nu \geq m_L(p)$ with equality only at the lower analogue. Neither extremal measure is admissible: its support is a half-line, whereas a pullback of Q through an increasing homeomorphism of $\mathbb{R}$ has support $\mathbb{R}$ because Q does. Conversely, with $\eta \in (0, 1]$, the mixtures $\nu_\eta := (1 - \eta)\nu^* + \eta F_2$ satisfy $p\nu_\eta \leq (1 - \eta)F_2 + \eta pF_2 \leq F_2$, are atomless with support $\mathbb{R}$, and have means filling $[\mathbb{E}_{F_2} Y, m_U(p))$; the lower analogue fills the rest of the

open interval. For each such ν the increasing rearrangement $c' := Q^{-1} \circ F_\nu$ is a continuous, strictly increasing map of $\mathbb{R}$ onto $\mathbb{R}$, because both distribution functions are continuous and strictly increasing, so $c' \in \mathcal{M}$ and $(c'^{-1})_\# Q = \nu$; thus $c' \in \mathcal{C}_{ID}$ by Theorem 1, and $h^S = \mathbb{E}[Y_2 \mid S] - \int y d\nu$ because $\mathbb{E}[Y_2 \mid S]$ is observed and finite. The distribution-function bounds are $\nu(-\infty, y] \leq F_2(y)/p$ and $\nu(y, \infty) \leq (1 - F_2(y))/p$. $\square$

The closure of the interval is the identification region of Horowitz and Manski (1995) [Proposition 1A, p. 285; Corollary 4.1, eq. (14), p. 291] for the mean of one component of a mixture with a known mixing proportion: here the refreshment distribution is the mixture, the survivors are the component of known proportion p , and the attriters' latent distribution is the free component. Its width is also the width of the trimming bounds of Lee (2009) at trimming proportion $1 - p$; Lee (2009, Appendix, Lemma 1, pp. 1097–1098) states the mixture result and attributes its proof to the corollary of Horowitz and Manski. The trimmed-mean interval is therefore an established result. What Corollary 6 adds is the statement for the map class $\mathcal{M}$ —interior points are attained, endpoints are not—and its use as the benchmark against which the location restriction is judged. For a Gaussian F_2 with standard deviation σ the width is $w(p) = 2\sigma \phi(\Phi^{-1}(p))/p$: $.70\sigma$ at $p = .8$ and 1.29σ at $p = .6$. A location shift h' places the survivors' latent mean at $\mathbb{E}[Y_2 \mid S] - h'$, so the location set of the next theorem is contained in the closure of this interval; under the logistic-selection example of Section 3.1 it has width at most $\beta\sigma^2 - .3\sigma$ for $\beta = .3$, $\sigma = 1$ —and under outcome-independent attrition with tails thinner than exponential it is a point. In the design of Simulation 1 the location sets have widths between 0 and $.26\sigma$ at $p = .8$, against $.70\sigma$ for an unrestricted map: the shape restriction, not the retention rate, does most of the work.

Remark 4. The mean bounds do not depend on the conditioning model. The necessity half of Corollary 6 uses only that the survivors are a sub-population of known proportion p of a population whose latent distribution the refreshment sample reveals; it never uses that conditioning is a deterministic increasing map. The interval $[m_L(p), m_U(p)]$ for the survivors' latent mean, and hence the corresponding interval for h^S , therefore bounds these quantities under any conditioning mechanism—stochastic, non-monotone, item-specific—and in a model rich enough to carry an arbitrary sub-population's latent distribution to the observed

reported one, as stochastic models are, the bounds are sharp and can be attained. With atoms the extremal sub-populations may split an atom, and the endpoints are the values of the linear programme $\max$ (or $\min$) $\int y d\nu$ over $0 \leq \nu \leq F_2/p$ with $\nu(\mathbb{R}) = 1$. For a binary item with population mean π_2 this gives $m^S \in [\max\{0, (\pi_2 - 1 + p)/p\}, \min\{1, \pi_2/p\}]$, an interval of width $(1 - p)/p$ whenever $1 - p \leq \pi_2 \leq p$. The bounds of Das, Toepoel, and Soest (2011) for a binary item have width $1 - p$: theirs is the *population* conditioning effect, whose unobserved part is the attriters' conditioned answers, ours the *survivor* effect, whose unobserved part is the survivors' latent mean. The two are different sharp answers to different questions, and the survivor interval is the wider because it divides the same unobserved mass $1 - p$ by the smaller denominator p . The location set of Theorem 3 and the mass criterion of Corollary 4, by contrast, do use the deterministic model, which is where their extra power comes from.

Restricting $\mathcal{M}$ to location shifts $c(y) = y + h$ turns the funnel into an interval and Theorem 1 into a bounds theorem.

Theorem 3 (Sharp bounds). *Under the conditions of Theorem 1 with $c(y) = y + h$, and assuming the refreshment density f_2 is unimodal with an upper semicontinuous version (for example, continuous),*

$$\mathcal{H}_{ID} = \left\{ h' : p q(y + h') \leq f_2(y) \text{ a.e.} \right\} = [h_L, h_U] \ni h,$$

where q is the density of Q : the identified set is a closed interval containing h , and it is sharp, every $h' \in \mathcal{H}_{ID}$ being attained by the explicit completion of Lemma 1. Writing $\Psi(h') := \text{ess sup}_y p q(y + h')/f_2(y)$, the endpoints are the infimum and supremum of $\{h' : \Psi(h') \leq 1\}$. The interval is nondegenerate exactly when $\Psi(h') \leq 1$ for some $h' \neq h$; Corollary 2(iii) gives a sufficient condition for a two-sided neighbourhood of h . It may be a single point at interior retention: with $f_2 = N(0, 1)$, $q = N(-1, 1)$ and $p = .5$ it is $\{-1\}$, and with $f_2 = N(0, 1)$, $q = N(0, \sigma^2)$ and $p = \sigma < 1$ it is $\{0\}$ although the stayers' latent density has strictly lighter tails than f_2 on both sides (the example following Corollary 2). Lighter tails of the stayers' latent density therefore do not by themselves make the interval nondegenerate. The interval is contained in the closure of the unrestricted-map interval of Corollary 6.

Proof. From Theorem 1, the location family makes $Q_{c'}$ a shift of Q and the domination constraint becomes the displayed density inequality. Rewriting it with $z = y + h'$, the constraint is equivalent to

$$f_2(z - h') \geq p q(z) \quad \text{for a.e. } z.$$

Interval. If $h'_1 < h'_2$ are feasible and $h'_1 < h' < h'_2$, then for a.e. z both $f_2(z - h'_1) \geq p q(z)$ and $f_2(z - h'_2) \geq p q(z)$ hold; $z - h'$ lies between $z - h'_2$ and $z - h'_1$, and a unimodal density at a point between two others is at least the smaller of its values there, so $f_2(z - h') \geq p q(z)$. *Closed.* If feasible $h'_n \rightarrow h'$, then outside the countable union of the exceptional null sets $f_2(z - h'_n) \geq p q(z)$ for every n , and upper semicontinuity gives $f_2(z - h') \geq \limsup_n f_2(z - h'_n) \geq p q(z)$. The set is nonempty because it contains the true h . (Log-concavity of f_2 is sufficient a fortiori; no condition on q is needed.) Sharpness of every point in it is Lemma 1. The characterization of the endpoints and of nondegeneracy restates the definition, and the Gaussian examples are computed in the example following Corollary 2 and in Section 12. Containment follows because a shift h' fixes the survivors' latent mean at $\mathbb{E}[Y_2 \mid S] - h'$, which must lie in $[m_L(p), m_U(p)]$ by the necessity half of Corollary 6. $\square$

Remark. Relation to Lee bounds. Lee (2009) bounds the average effect of a training programme on wages for those whose wages would be observed under either treatment assignment, by trimming the observed outcome distribution of the treated group from either tail by the proportion $p_0 = [\Pr(S = 1 \mid D = 1) - \Pr(S = 1 \mid D = 0)] / \Pr(S = 1 \mid D = 1)$, under random assignment and the monotonicity restriction $S_1 \geq S_0$ (Lee 2009, Proposition 1a, p. 1083). In his application the observation indicator is employment; the method is stated for sample selection in general, and his conclusion notes its application to survey nonresponse and attrition (Lee 2009, 1097). Corollary 6 trims the *refreshment* distribution at the retention rate to bound the survivors' latent mean. The two share a geometry—a sub-measure of known mass must fit under a fixed measure, and the extremal fits are at the tails—but differ in estimand and in the assumptions that deliver the mixture: the unknown bounded here is the *survivors'* latent distribution, whose mixture with the attriters' latent distribution is the refreshment distribution; Lee's is the outcome distribution of treated units who would be

observed under either assignment, and the mixture is delivered by independence and monotonicity. The common geometry does not make the estimands or the selection assumptions the same. The containment statement of Theorem 3 says what the location restriction adds to the unrestricted-map interval.

5.1 Shrinking the funnel with covariates

Covariates observed for the whole cohort and in the refreshment sample shrink the identified set through stratification; this is what the covariate-standardized designs of Section 7 exploit.

Proposition 1 (Covariate stratification). *Let X be a discrete entry covariate observed for the whole cohort and in the refreshment sample, let structures carry X (so that F is a law on (X, Y_1, Y_2^*) and π may depend on X), and suppose the conditioning map does not vary with X . Write $p(x)$, $Q(\cdot | x)$, and $F_2(\cdot | x)$ for the retention rate, the stayers' reported-outcome distribution, and the refreshment distribution within $X = x$. Then the identified set is*

$$\mathcal{C}_{ID}(X) = \bigcap_x \{c' \in \mathcal{M} : p(x) Q_{c'}(\cdot | x) \leq F_2(\cdot | x)\} \subseteq \mathcal{C}_{ID},$$

it is sharp, and in the location family $\mathcal{H}_{ID}(X) = \bigcap_x \mathcal{H}_{ID}(x)$. The inclusion is strict exactly when some map satisfying the pooled constraint violates the constraint of some stratum; this can happen when the strata differ only in their retention rates. If the map is allowed to vary with X , the identified set is the product of the stratum sets.

Proof. Necessity within each stratum is Theorem 1 applied conditionally on $X = x$: the stratum's observables are $(F_1(\cdot | x), F_2(\cdot | x), G(\cdot | x), p(x))$, and a structure that rationalizes the pooled data rationalizes each stratum's. Sufficiency: for c' in the intersection, complete each stratum by Lemma 1 and glue, $F' := \sum_x \Pr(X = x) F'(\cdot | x)$ and $\pi'(y_1, y_2, x) := \pi'_x(y_1, y_2)$. The inclusion: the stayers' sub-measure is the mixture $pQ = \sum_x \Pr(X = x) p(x) Q(\cdot | x)$ and the pullback commutes with mixing, so summing the stratum inequalities with weights $\Pr(X = x)$ gives $pQ_{c'} \leq \sum_x \Pr(X = x) F_2(\cdot | x) = F_2$. The strictness statement restates the definition of the two sets, and the retention-only case is the example below. The product statement is immediate. $\square$

Two readings of the proposition should be kept apart. If selection is *known* to be independent of the latent outcome within strata—missing at random given X —then the stayers’ latent distribution within each stratum equals $F_2(\cdot \mid x)$, and Corollary 1 applied within a stratum point-identifies the map by quantile alignment. The data cannot reveal that this is so: the observables generated under selection on X alone are also generated by outcome-dependent selection within strata, and without imposing the assumption the covariate shrinks the identified set only through the intersection. A covariate can tighten the set even when it predicts retention and nothing else. Take two equally prevalent strata with identical Laplace distributions of rate one, $F_2(\cdot \mid x) = Q(\cdot \mid x)$, the true map the identity, and retention .8 and .2. The pooled retention rate is .5 and the pooled location set is $[-\log 2, \log 2]$ (Corollary 3(ii)); the stratum sets are $[-\log 1.25, \log 1.25]$ and $[-\log 5, \log 5]$, whose intersection $[-\log 1.25, \log 1.25]$ is strictly smaller, because shifts with $\log 1.25 < |\varepsilon| \leq \log 2$ satisfy the pooled constraint but violate the high-retention stratum’s.

6 Two Refreshments: Point Identification of the Path

A single refreshment bounds a single contrast. Panels with *multiple* refreshments observe multiple tenure configurations in common periods, and the bounds question becomes a rank question. This section gives the full linear-system treatment.

6.1 The moment system

Entry cohorts $e_1 < e_2 < e_3$ (JLPS: 2007, 2011, 2019), tenure $s = t - e + 1$. A *refreshment episode* r is a wave at which a new cohort enters alongside surviving older cohorts: episode 1 ($t = 2011$) generates tenures $\{5, 1\}$, episode 2 ($t = 2019$) generates $\{13, 9, 1\}$. Outcomes are standardized item by item; for a target item, write the survivor mean at episode r and tenure s as

$$m_r(s) = \mu_r + g(e) + \tau(s) + \beta_r(s),$$

where μ_r is the period effect, $g(e)$ the entry-cohort effect, $\tau(s)$ the conditioning path with the entry normalization $\tau(1) = 0$, and $\beta_r(s)$ the *selection functional*: the standardized difference

between the surviving and full entry cohort at tenure s , with $\beta_r(1) = 0$ because a refreshment cohort at entry is an unselected draw. Within an episode, period effects cancel in contrasts and cohort is a function of tenure, so the estimable moments are the *entry contrasts*

$$C_r(s) = m_r(s) - m_r(1) = \Delta g_r(s) + \tau(s) + \beta_r(s),$$

one per experienced tenure per episode: $C_1(5), C_2(9), C_2(13)$. Negative-control items—time-invariant facts reported once at entry, so $\tau^N \equiv 0$ by construction, and reported *before* any panel selection, so their survivor means at t equal the entry values of survivors—yield the parallel moments

$$N_r(s) = \Delta g_r^N(s) + \beta_r^N(s).$$

Two transport assumptions link the systems, both inherited from the recovery conditions of the companion identification paper and both partially testable here:

(A1) Cohort overlap and comparability. Each refreshment targets the same birth cohorts as the incumbents on a common support (in the Japanese panel, the age-matched refreshment of 2011), *and* on that support entrants and incumbents are drawn from the same population up to sampling, so that $\Delta g_r(s) = \Delta g_r^N(s) = 0$ for durable items. Overlap of birth-year support is necessary for the second clause but does not imply it: comparability is a separate assumption about sampling frames, nonresponse at entry, mode, and the coding of items. (Off support, Δg is bounded rather than zero and the conclusions become interval-valued; see Corollary 7.)

(A2) Selection transport. For target item j , $\beta_r^j(s) = \Gamma_j \beta_r^N(s)$ with loading Γ_j known or bounded: the item-level selection differential is the common survivor-composition shift measured on the negative controls, scaled by the item’s loading on it. ($\Gamma_j = 1$ for all standardized items is a leading case; a bounded $\Gamma_j \in [\Gamma_L, \Gamma_U]$ gives intervals.) (A2) restricts the joint selection of the target item and the battery; it is not implied by (A1), and a small negative-control contrast does not by itself show that the target item’s selection bias is small.

6.2 The theorem

Theorem 4 (Point identification with two refreshments: the role of the negative-control battery). *Under (A1)–(A2), in a design with three entry cohorts and two age-matched refreshment episodes that generate experienced-cohort contrasts at the tenure pairs $(r, s) \in \{(1, 5), (2, 9), (2, 13)\}$ (the episode structure of the Japanese panel, whose second refreshment is however generational; see Corollary 7 for that case):*

- (a) (Sufficiency.) *If every episode that contributes an experienced-cohort contrast carries a negative-control battery and the loading Γ is known, then $\tau(s)$ is point-identified at every tenure s at which an experienced cohort is observed at a refreshment episode:*

$$\tau(s) = C_r(s) - \Gamma N_r(s), \quad (r, s) \in \{(1, 5), (2, 9), (2, 13)\}.$$

No stationarity of selection across episodes is required: each episode’s selection functional is measured within that episode. If the loading is known only to lie in an interval, $\Gamma \in [\Gamma_L, \Gamma_U]$, then, because the loading is common to all pairs (r, s) , the pairs restrict it jointly: the identified set for the loading is

$$\Gamma^* := \{ \Gamma \in [\Gamma_L, \Gamma_U] : C_r(s) - \Gamma N_r(s) \in \mathcal{H}_r(s) \text{ for every observed } (r, s) \},$$

with $\mathcal{H}_r(s)$ the location set of Theorem 3 at episode r expressed in the units of the moment system (if a tenure s were observed at two episodes, Γ would also have to make $C_r(s) - \Gamma N_r(s)$ equal across them, and the relevant set would be the intersection of Corollary 5), and the identified set for the vector $(\tau(s))_s$ is $\{(C_r(s) - \Gamma N_r(s))_{(r,s)} : \Gamma \in \Gamma^\}$. Intersecting each pair’s interval $\{C_r(s) - \Gamma N_r(s) : \Gamma \in [\Gamma_L, \Gamma_U]\}$, of width $(\Gamma_U - \Gamma_L) |N_r(s)|$, with its own $\mathcal{H}_r(s)$ gives an outer bound, which can be strictly larger. The set is a point when $N_r(s) = 0$ at every pair or when the loading is known.*

- (b) (Necessity.) *Without a negative-control battery at episode r (and without stationarity), $\tau(s)$ and $\beta_r(s)$ enter the observable moments of that episode only through their sum: for any ϵ , setting $\tilde{\tau}(s) = \tau(s) + \epsilon$ and $\tilde{\beta}_r(s) = \beta_r(s) - \epsilon$ reproduces all of them. The perturbation is attainable by an admissible selection process whenever the corre-*

sponding location shift $h + \epsilon$ lies in the location set of Theorem 3 at every episode at which tenure s is observed and none of those episodes carries a battery used under (A2). If those sets contain $h + \epsilon_0$ for some $\epsilon_0 \neq 0$ and are intervals (as under unimodality), every ϵ between 0 and ϵ_0 is attainable; a one-sided set permits perturbations of one sign only. For every attainable $\epsilon \neq 0$, $\tau(s)$ is not point-identified, and the non-identification is not a moments-only artefact: by Theorem 2 the perturbed structure can be chosen to reproduce the entire observed distribution of every cohort involved—every response pattern’s joint distribution of reports across all waves, the entry-wave battery, and the retention pattern—together with the refreshment distributions, so that (A1) is preserved and no functional of the data distinguishes $\tau(s)$ from $\tau(s) + \epsilon$. The condition is not automatic. Under attrition independent of the latent outcome and a refreshment density with tails thinner than exponential, the location set is $\{h\}$ (Corollary 3(i)); under outcome-dependent attrition it can also be $\{h\}$ (the example following Corollary 2). Necessity therefore requires a nondegenerate location set, not merely outcome-dependent attrition. Cross-episode stationarity ($\beta_1 = \beta_2 = \beta$) does not substitute when the episodes’ tenure sets are disjoint—as in the Japanese panel, $\{5\} \cap \{9, 13\} = \emptyset$ —because the common function $\beta(\cdot)$ is never evaluated twice at the same argument, so the constraint binds nothing; when a tenure is repeated it adds, in the location family, a testable restriction without shrinking the identified set (Corollary 5). Restrictions that link cohorts or waves beyond (A1)–(A2), such as an additive cell-mean structure across cohorts, must be checked against the perturbation separately.

- (c) (Overidentification.) With a K -item negative-control battery whose loadings are known, each pair (r, s) delivers K estimates of the single factor $\beta_r^N(s)$, giving $K - 1$ testable restrictions per pair ($3(K - 1)$ in the configuration of (a); $K = 23$ in the Japanese panel)—the battery-dispersion placebo test. Under the additional hypothesis of stationarity in tenure ($\beta_r(s) = \beta(s)$ for a common β), no further restriction is testable in that tenure configuration (by the disjointness in (b)); a design with a repeated tenure across episodes—for example, refreshments eight waves apart in a panel observed at tenure 9 in both—adds one direct stationarity test per repeated tenure.

Proof.

- (a) Substitute (A1) into the moment equations: $C_r(s) = \tau(s) + \Gamma \beta_r^N(s)$ and $N_r(s) = \beta_r^N(s)$.

The system, stacked over $(r, s) \in \{(1, 5), (2, 9), (2, 13)\}$, is triangular: each $N_r(s)$ identifies its $\beta_r^N(s)$, and each $C_r(s)$ then identifies its $\tau(s)$ by subtraction; the coefficient matrix is the identity on $(\beta_1^N(5), \beta_2^N(9), \beta_2^N(13))$ stacked with a unit-diagonal block on $(\tau(5), \tau(9), \tau(13))$. With Γ unknown, every Γ in Γ^* is attained: fix it and set $\tilde{\tau}(s) = C_r(s) - \Gamma N_r(s)$, which lies in $\mathcal{H}_r(s)$ by definition of Γ^* . For each cohort, Theorem 2 applied with J^* the tenures at which that cohort meets a refreshment sample, and with the entry coordinate carrying the battery, gives a structure that reproduces all of the cohort's observables and the refreshment distributions while giving the survivors' latent mean at each such tenure the value implied by $\tilde{\tau}$; then $\tilde{\beta}_r(s) = C_r(s) - \tilde{\tau}(s) = \Gamma N_r(s)$, so (A2) holds with loading Γ , and (A1) holds because the population marginals are the refreshment distributions. Cohorts are sampled independently, so the per-cohort structures combine. Conversely, a Γ outside Γ^* implies, at some pair, a shift outside the location set, which Theorem 3 excludes. The outer bound drops the requirement that one Γ serve all pairs; the deterministic check (11) of Section 12 gives a case in which it is strictly larger.

- (b) The observable moments of episode r involve (τ, β_r) only through $C_r(s) = \tau(s) + \beta_r(s)$ (given (A1)), so the map $(\tau, \beta_r) \mapsto$ moments factors through the sum, and the ϵ -perturbation leaves the sum invariant. If the location shift $h + \epsilon$ lies in the set of Theorem 3 at every episode observing tenure s , then for each cohort observed at tenure s at a refreshment episode, Theorem 2 with J^* its refreshment tenures, the candidate map at s shifted by ϵ and the true maps at its other refreshment tenures, and with the entry coordinate carrying the negative-control battery, produces a structure that reproduces every observed pattern sub-measure, the entry marginal (hence every negative-control moment and every entry-wave selection differential), and the refreshment marginals; under it the survivors' latent mean at that episode moves by $-\epsilon$, so $\tilde{\beta}_r(s) = \beta_r(s) - \epsilon$ and $\tilde{\tau}(s) = \tau(s) + \epsilon$ while $C_r(s)$ and $N_r(s)$ are unchanged. Under unimodality the location sets are intervals containing h (Theorem 3), which gives the statement about

ϵ between 0 and ϵ_0 . For the stationarity clause: the hypothesis $\beta_1 = \beta_2$ constrains the function $\beta(\cdot)$ only at arguments where both episodes evaluate it; with disjoint tenure sets the constraint set is vacuous, so the perturbation above again reproduces all moments; with a repeated tenure, Corollary 5.

- (c) Under (A2) the K negative-control moments at (r, s) are $N_r^{(k)}(s) = \lambda_k \beta_r^N(s)$, $k = 1, \dots, K$, with known loadings λ_k : a one-factor system with K observations and one unknown, hence $K - 1$ restrictions per (r, s) , testable as dispersion of the battery around its common factor. The final clause follows from the argument in (b). $\square$

$\square$

Corollary 7 (Generational refreshment). *Suppose episode r 's refreshment cohort shares no birth-cohort support with the incumbents, so (A1) fails between entrants and incumbents, but two or more incumbent cohorts at episode r do overlap. Then: (i) level contrasts against the entrants identify only the composite $\Delta g_r(s) + \tau(s) + \beta_r(s)$; after negative-control correction under (A2) with loading Γ , $C_r(s) - \Gamma N_r(s) = \tau(s) + \{\Delta g_r(s) - \Gamma \Delta g_r^N(s)\}$, so $\tau(s)$ is interval-identified, with width $2M$, under a bound $|\Delta g_r(s) - \Gamma \Delta g_r^N(s)| \leq M$ on the residual cohort effect after transport; a bound on $|\Delta g_r(s)|$ alone does not suffice when $\Delta g_r^N(s)$ is unrestricted; (ii) the increment between incumbent tenures— $\tau(s) - \tau(s')$ for incumbent cohorts at tenures s, s' —remains point-identified on the incumbents' common support under (A1) between the two incumbent cohorts and (A2) with a known loading for their contrast (a bounded loading gives an interval), by applying Theorem 4's argument to the incumbent pair, with the negative-control contrast between the same two cohorts supplying $\beta_r^N(s) - \beta_r^N(s')$.*

Proof.

- (i) With disjoint support, $\Delta g_r(s)$ cannot be zeroed by (A1) and enters every entrant contrast, and $\Delta g_r^N(s)$ enters every negative-control contrast; substituting $\beta_r(s) = \Gamma \beta_r^N(s)$ gives the display, and the bound on the residual gives the interval. If only $|\Delta g_r(s)| \leq M$ is assumed, $\Gamma \Delta g_r^N(s)$ is free and the corrected contrast is unrestricted.
- (ii): restrict to the two incumbent cohorts (entered at e_1, e_2 , observed at episode r with tenures $s > s'$) on their overlapping birth years; under (A1) within this pair the

target contrast is $[\tau(s) - \tau(s')] + [\beta_r(s) - \beta_r(s')]$ and the same-pair negative-control contrast is $\beta_r^N(s) - \beta_r^N(s')$; subtract under (A2). $\square$

$\square$

Real refreshments come in both kinds. An *age-matched* refreshment replenishes the same birth cohorts (attrition repair); a *generational* refreshment extends the panel to a new cohort (coverage repair). Chadi (2021) distinguishes a different pair of entrant types—entrants induced by the data collector and “natural” entrants into a household panel—and uses their coexistence to separate attrition bias from participation-experience effects; the distinction here is between entrants whose birth years overlap the incumbents’ and entrants from a new birth cohort. The Japanese panel’s 2011 episode is of the first kind and its 2019 episode of the second—the 2019 entrants’ birth years (1987–1998) do not intersect the incumbents’ (1966–1986)—so under (A1)–(A2) the panel identifies $\tau(5)$ (episode 1, levels) and $\tau(13) - \tau(9)$ (episode 2, increments), while $\tau(13)$ ’s level rests on a bound on the residual cohort effect. Panels planning refreshments should know that the choice between the two kinds is a choice about what becomes identified.

Three comments. First, the practical content of (a) is that *the negative-control battery, together with (A2), and not the second refreshment per se, is what buys point identification*; the second refreshment’s contribution is to extend the set of tenures at which τ is recovered (from $\{5\}$ to $\{5, 9, 13\}$ when the second refreshment is age-matched, and to the increment $\tau(13) - \tau(9)$ when it is generational, as in the Japanese panel) and to double the placebo system. Second, (b) corrects a natural intuition—that stationary selection plus two refreshments should identify the path by differencing. It does not, in the configuration real panels have (disjoint tenures); the intuition is right only in designs engineered to repeat a tenure, and there only for the restriction, not for the set. This is the selection-side counterpart of the non-identification result in the companion identification paper: there, tenure–period–cohort collinearity; here, the sum structure $\tau + \beta_r$. Third, (c) turns the placebo battery of the companion empirical paper from a robustness check into an identifying instrument under (A2), and its empirical behaviour there—median $|d| = .031$ across 23 items, three rejections of which two are recall items that depend on the era of childhood (radio, telephone),

motivating the restriction of the battery to durable facts—is directly the test this theorem prescribes.

7 Constructive Designs: Restrictions That Identify the Survivors’ Mean Shift

Theorem 1 says what a refreshment sample identifies without restrictions on attrition; Theorem 4 says what a second refreshment and a negative-control battery add. Between the two lies applied practice, where a single refreshment sample is combined with a restriction on the attrition process, often stated in words, to obtain a point estimate: the survival-matching design of Halpern-Manners, Warren, and Torche (2017) is the best known. This section states such designs as restrictions within the theory. Each is a restriction on the selection functional that identifies the survivors’ mean shift; each is biased in a characteristic way when its restriction fails; and, because they rest on different restrictions, their disagreements can be used to check the compatibility of the restrictions. The designs are stated in the tenure notation used throughout, with the dose k of prior interviews written as $k = s - 1$.

7.1 Multi-wave setting and the selection functional

Specialize the multiwave setting of Section 4.1 to the cohort structure of the moment system (Section 6.1) under monotone attrition. A continuing cohort enters at wave e and is compared, at wave $t > e$, with a fresh cohort entering at t ; the continuing cohort’s tenure is $s = t - e + 1$ (dose $k = s - 1$). Write $R_{iw} \in \{0, 1\}$ for a completed interview at wave w and $S_{i,a:b} = \prod_{w=a}^b R_{iw}$ for survival through waves $a, \dots, b$. The decision governing $R_{i,w+1}$ is taken at wave w , after the interview at w , and may depend on the transient component ε_{iw} of that wave’s latent response; hence $S_{i,e:t} = 1$ reflects decisions taken at waves $e, \dots, t - 1$ and is not selected on ε_{it} , whereas a fresh cohort’s $S_{i,t:t+k} = 1$ reflects decisions at $t, \dots, t + k - 1$ and is selected on ε_{it} . This is the timing implemented in the simulation code and the package. The latent (unconditioned) response at t is Y_{it}^* ; the continuing survivors report $Y_{it} = c(Y_{it}^*)$, the fresh cohort reports Y_{it}^* . In the location-shift case $c(y) = y + h$, and we

write $h^S := \mathbb{E}[Y_t - Y_t^* \mid S_{e:t} = 1]$ for the mean shift among survivors, which reduces to h when the shift is homogeneous.

The object that every design must pin down is the *selection functional*

$$\delta_e(w) := \mathbb{E}[Y_w^* \mid S_{e:t} = 1, \text{ cohort } e] - \mathbb{E}[Y_w^* \mid \text{cohort } e], \quad w \in \{e, \dots, t\}, \quad (2)$$

the difference between eventual survivors and the whole entry cohort on the latent response at wave w . At $w = t$ it is the raw-scale version of $\beta_r(s)$ in Section 6.1; at $w = e$ it is observable, because every entrant reports at entry and entry reports are unconditioned. The naive contrast between continuing survivors and the fresh cohort decomposes, under refreshment validity, as

$$C := \mathbb{E}[Y_t \mid S_{e:t} = 1] - \mathbb{E}[Y_t^* \mid \text{fresh}] = h^S + \delta_e(t), \quad (3)$$

so that identifying h^S is identifying $\delta_e(t)$; Theorem 1 is the statement that the data at t bound $\delta_e(t)$ without pinning it, and Theorem 2 the statement that the cohort's own earlier waves do not pin it either. Every design below therefore rests on a restriction that links $\delta_e(t)$ to something observed.

We use the following restrictions. **B1** is refreshment validity together with cohort overlap (A1): the fresh cohort's latent distribution at t equals the continuing cohort's population latent distribution at t on a common support.

- **B2 (survival-matched attrition exchangeability).** $\mathbb{E}[Y_t^* \mid S_{e:t} = 1, \text{ cont}] = \mathbb{E}[Y_t^* \mid S_{t:t+k} = 1, \text{ fresh}]$: fresh respondents who go on to survive the same number of waves as the continuing survivors have the same latent mean.
- **B3 (stationary, symmetric attrition).** $\mathbb{E}[Y_t^* \mid S_{e:t+1} = 1, \text{ cont}] = \mathbb{E}[Y_t^* \mid S_{t:t+k+1} = 1, \text{ fresh}]$: when both arms are selected by exactly $k + 1$ response decisions, one of them taken at t , the selected latent means coincide. Sufficient primitive conditions: the per-wave response decision $\pi_w(X_i, U_i, \varepsilon_{iw})$ has the same form at every wave, does not depend on tenure, and the transient states ε_{iw} are exchangeable across waves given (X_i, U_i) .

- **B4 (time-invariant selection differential).** $\delta_e(t) = \delta_e(e)$: the survivors' advantage on the latent response is the same at the comparison wave as at entry. This is (A2) of Section 6.1 with the negative control replaced by the item's own entry-wave report and loading $\Gamma = 1$; it is a parallel-trends condition on the latent outcome between survivors and the full cohort.

Three levels should be kept apart. B2–B4 are restrictions on *means* of the latent outcome, and what they identify is the mean shift h^S (or h^{S+}); they do not identify the conditioning map c itself, nor the selection process. The distributional versions of survival matching and symmetric matching—in which the whole latent distribution of the matched fresh arm is taken as that of the survivors, so that c is recovered by the quantile alignment of Corollary 1—require the corresponding restrictions to hold *in distribution*, which is stronger than B2 or B3 and which we do not test. And no mean restriction pins down a selection function: many selection processes are compatible with B2, B3, or B4, which is exactly why the identified set of Theorem 1, rather than any particular selection model, is the right object to compare the designs against.

B2–B4 are alternatives, not a hierarchy. B2 needs k waves of follow-up on the fresh cohort; B3 needs $k + 1$; B4 needs none, but needs the item at the cohort's entry wave with the same coding.

7.2 Three designs and what each identifies

Proposition 2 (Survival matching). *Under B1 and B2, h^S is point identified: $h^S = \mathbb{E}[Y_t \mid S_{e:t} = 1, \text{ cont}] - \mathbb{E}[Y_t^* \mid S_{t:t+k} = 1, \text{ fresh}]$. If B2 holds in distribution (the two conditional distributions of Y_t^* coincide), the whole map c is point identified, $c = Q^{-1} \circ F_{Y_t^* \mid S_{t:t+k}=1, \text{ fresh}}$, and the identified set $\mathcal{C}_{ID}$ of Theorem 1 collapses to $\{c\}$.*

Proposition 3 (Symmetric survival matching). *Under B1 and B3, the mean shift among survivors who also respond at $t + 1$, $h^{S+} := \mathbb{E}[Y_t - Y_t^* \mid S_{e:t+1} = 1]$, is point identified: $h^{S+} = \mathbb{E}[Y_t \mid S_{e:t+1} = 1, \text{ cont}] - \mathbb{E}[Y_t^* \mid S_{t:t+k+1} = 1, \text{ fresh}]$; with B3 in distribution, c is point identified on that subpopulation. The two survivor estimands differ by a composition term, $h^S = r h^{S+} + (1 - r) h^{S-}$ with $r = \Pr(R_{t+1} = 1 \mid S_{e:t} = 1)$ and h^{S-} the shift among*

survivors who do not respond at $t + 1$.

Proposition 4 (Entry-wave correction (the Das–Toepoel–van Soest functional)). *Under B1 and B4, h^S is point identified from the continuing cohort’s own entry wave and the fresh cohort’s first wave:*

$$h^S = \mathbb{E}[Y_t \mid S_{e:t} = 1, \text{ cont}] - \underbrace{(\mathbb{E}[Y_e \mid S_{e:t} = 1, \text{ cont}] - \mathbb{E}[Y_e \mid \text{cont}])}_{\delta_e(e)} - \mathbb{E}[Y_t^* \mid \text{fresh}]. \quad (4)$$

The identified interval $\mathcal{H}_{ID}$ of Theorem 3 collapses to the point $C - \delta_e(e)$. The design uses no follow-up of the fresh cohort, is feasible for every $s \leq t$, and admits a covariate-standardized version under the conditional restriction $\delta_e(t \mid x) = \delta_e(e \mid x)$ for entry covariates x . The observable combination in Equation 4— $\mathbb{E}[Y_t \mid S] - \mathbb{E}[Y_e \mid S] + \mathbb{E}[Y_e] - \mathbb{E}[Y_t^* \mid \text{fresh}]$ —is the one Das, Toepoel, and Soest (2011, Assumption 3) use for a binary item, and the proposition should be read as a reinterpretation of their estimator rather than as a new one. The two differ in target and in restriction. Their stationary-attrition-bias assumption concerns the conditioned outcome and identifies a population conditioning effect; B4 concerns the unconditioned latent outcome and identifies the effect among survivors, h^S . Under a homogeneous shift the two coincide; under heterogeneous effects they do not, and a design can satisfy one restriction while violating the other. Simulation 3 (Appendix) includes a regime in which the survivor and population targets differ.

Proof. All three follow from Equation 3. Under B1 the fresh cohort’s mean of Y_t^* (or of its survival-matched subset) is an unbiased proxy for the corresponding continuing-cohort latent mean whenever the restriction identifies the survivors’ latent mean: under B2 it is the fresh k -wave survivors’ mean, under B3 the fresh $(k+1)$ -wave survivors’ mean (with the continuing arm restricted to $S_{e:t+1} = 1$), under B4 the population mean plus $\delta_e(e)$, which is observed because entry reports are unconditioned and available for every entrant. Subtracting the proxy from the survivors’ reported mean leaves h^S (or h^{S+}). The distributional statements replace means by distributions and apply Corollary 1: once the stayers’ latent distribution is known, c is the increasing rearrangement transporting it to Q . The composition identity in Proposition 3 is the law of total expectation over R_{t+1} among survivors. $\square$

The three propositions are one argument seen from three sides: each restriction identifies the relevant survivor mean without uniquely determining the selection function, by asserting where in the data the survivors' latent mean can be read off. Survival matching reads it off the future of the fresh cohort; symmetric matching does the same after equalizing the number of selection decisions; the entry-wave correction reads it off the past of the continuing cohort. Theorem 4 is the same idea with a battery of items whose h is zero by construction in place of the item's own entry wave; the entry-wave correction is its single-item version, available for every item but with no overidentification.

7.3 Failure modes

The three designs rest on different restrictions, so they fail under different violations. The bias identities below are exact; the directions require a model of how traits, transient states and response decisions are related.

Proposition 5 (Biases under violation). *Let $\Delta_{SM} := \mathbb{E}[Y_t^* \mid S_{e:t} = 1, \text{ cont}] - \mathbb{E}[Y_t^* \mid S_{t:t+k} = 1, \text{ fresh}]$, $\Delta_{SSM} := \mathbb{E}[Y_t^* \mid S_{e:t+1} = 1, \text{ cont}] - \mathbb{E}[Y_t^* \mid S_{t:t+k+1} = 1, \text{ fresh}]$, and $\eta := \delta_e(t) - \delta_e(e)$. Then, under B1,*

$$\hat{h}_{SM} \rightarrow h^S + \Delta_{SM}, \quad \hat{h}_{EC} \rightarrow h^S + \eta, \quad \hat{h}_{SSM} \rightarrow h^{S+} + \Delta_{SSM}.$$

Suppose in addition model (M): the latent response is $Y_{iw}^ = \mu_w + \lambda'_w A_i + \varepsilon_{iw}$, with a vector A_i of person-level traits distributed identically in the two cohorts and transient innovations ε_{iw} that are independent and identically distributed across persons and waves, have mean zero, and are independent of A_i ; and the response decision taken at wave w continues participation with probability $\pi_w(A_i, \varepsilon_{iw})$, the decisions being independent across waves given (A_i, ε_i) . Then:*

- (i) (State-dependent, stationary attrition.) *If $\lambda_w \equiv \lambda$ and $\pi_w \equiv \pi$ at every wave of both cohorts, then $\Delta_{SSM} = 0$ and*

$$\Delta_{SM} = \eta = -\mathbb{E}[\varepsilon_t \mid S_{t:t+k} = 1, \text{ fresh}];$$

if $\pi(a, \cdot)$ is nondecreasing for every a (continuation more likely at higher transient values), $\Delta_{SM} = \eta \leq 0$, and if it is nonincreasing, $\Delta_{SM} = \eta \geq 0$. Survival matching and the entry-wave correction carry the same bias, and symmetric matching is unbiased for h^{S+} .

(ii) (Non-stationary trait selection.) If $\lambda_w \equiv \lambda$ and $\pi_w(a, \varepsilon) = \pi_w(a)$ does not depend on ε , then $\eta = 0$, while $\Delta_{SM} = \lambda' \{ \mathbb{E}[A \mid S_{e:t} = 1, \text{cont}] - \mathbb{E}[A \mid S_{t:t+k} = 1, \text{fresh}] \}$ and $\Delta_{SSM} = \lambda' \{ \mathbb{E}[A \mid S_{e:t+1} = 1, \text{cont}] - \mathbb{E}[A \mid S_{t:t+k+1} = 1, \text{fresh}] \}$: the entry-wave correction is unbiased, and survival and symmetric matching are biased whenever these trait contrasts are nonzero, that is, whenever selection makes the survivors' mean of $\lambda' A$ differ between the continuing cohort's selection waves and the fresh cohort's.

(iii) (Drift in the trait loading.) If $\pi_w(a, \varepsilon) = \pi(a)$ at every wave of both cohorts but $\lambda_t \neq \lambda_e$, then $\Delta_{SM} = \Delta_{SSM} = 0$ and $\eta = (\lambda_t - \lambda_e)' \{ \mathbb{E}[A \mid S_{e:t} = 1, \text{cont}] - \mathbb{E}[A \mid \text{cont}] \}$: only the entry-wave correction is biased; this is the failure of the parallel-trends content of $B4$.

Outside (M) the directions are not implied. With symmetric innovations and continuation depending on ε_t^2 , selection leaves the mean of ε_t unchanged among survivors; and with serially dependent innovations, survivors selected on earlier innovations are also selected on the current one.

Proof. The identities: substitute the proxies of Proposition 2–Proposition 4 into Equation 3; the SM proxy misses the continuing survivors' latent mean by Δ_{SM} , the SSM proxy misses the corresponding mean of the survivors through $t + 1$ by Δ_{SSM} , and the EC proxy misses by $\delta_e(t) - \delta_e(e)$. Under (M) the common term μ_t cancels from Δ_{SM} , Δ_{SSM} and η . The continuing survivors $S_{e:t} = 1$ have passed the k decisions taken at waves $e, \dots, t - 1$, and the fresh survivors $S_{t:t+k} = 1$ the k decisions taken at $t, \dots, t + k - 1$.

(i) With stationary π and λ and identically distributed innovations, the joint law of $(A, \text{decisions})$ is the same for the k decisions of the two arms, so $\mathbb{E}[A \mid S_{e:t}, \text{cont}] = \mathbb{E}[A \mid S_{t:t+k}, \text{fresh}]$. The innovation ε_t is independent of A and of the continuing cohort's decisions before t , so $\mathbb{E}[\varepsilon_t \mid S_{e:t}, \text{cont}] = 0$, whereas the fresh cohort's first

decision is taken at t and depends on ε_t ; hence $\Delta_{SM} = -\mathbb{E}[\varepsilon_t \mid S_{t:t+k}, \text{fresh}]$. For the entry-wave correction, $\delta_e(t) = \lambda' \{\mathbb{E}[A \mid S_{e:t}] - \mathbb{E}A\}$ because ε_t is independent of the decisions before t , while $\delta_e(e) = \lambda' \{\mathbb{E}[A \mid S_{e:t}] - \mathbb{E}A\} + \mathbb{E}[\varepsilon_e \mid S_{e:t}]$ because the first decision of the continuing cohort is taken at e and depends on ε_e : the survivors are selected on the *entry* innovation, not on the current one. So $\eta = -\mathbb{E}[\varepsilon_e \mid S_{e:t}, \text{cont}]$, and since the continuing cohort's survivors pass k decisions, the first depending on ε_e , exactly as the fresh cohort's pass k decisions, the first depending on ε_t , the two expectations are equal: $\eta = \Delta_{SM}$. For the sign, condition on $A = a$: the other decisions are independent of the relevant innovation given a , so $\mathbb{E}[\varepsilon_t 1\{S\} \mid a]$ has the sign of $\text{Cov}(\varepsilon_t, \pi(a, \varepsilon_t))$, which is nonnegative when $\pi(a, \cdot)$ is nondecreasing, by the covariance inequality for monotone functions of one variable, and nonpositive when it is nonincreasing. For Δ_{SSM} , both arms pass $k+1$ decisions, exactly one of them taken at t and depending on ε_t ; because the innovations are exchangeable across waves and π is stationary, the position of that decision among the $k+1$ does not change the joint law of (A, ε_t) given survival, so the two selected latent means coincide.

(ii) Decisions do not depend on innovations, so $\mathbb{E}[\varepsilon_w \mid S] = 0$ for every w and both arms, $\delta_e(t) = \delta_e(e) = \lambda' \{\mathbb{E}[A \mid S_{e:t}] - \mathbb{E}A\}$, and $\eta = 0$; the expressions for Δ_{SM} and Δ_{SSM} are the trait parts of the definitions.

(iii) Both arms of SM and of SSM pass the same number of decisions under the same π , so their trait laws coincide and the common loading λ_t at the comparison wave cancels; the innovations do not enter the decisions; and $\delta_e(t) - \delta_e(e) = (\lambda_t - \lambda_e)' \{\mathbb{E}[A \mid S_{e:t}] - \mathbb{E}A\}$.

The two examples in the last paragraph are immediate. $\square$

$\square$

Under (M) the two failures that matter in practice bias different subsets of the three estimators: stationary state dependence biases survival matching and the entry-wave correction equally and leaves symmetric matching unbiased; non-stationary trait selection biases the two matching designs and leaves the entry-wave correction unbiased. Section 8 turns their disagreements into compatibility checks. Outside (M) the directions are not implied; Simu-

lation 3 illustrates them in one parametrization.

7.4 Increments, generational refreshments, and cohort-level nuisances

Two further constructions transfer directly. First, when two *continuing* cohorts drawn from the same birth cohorts are observed at a common wave t with tenures $s_A > s_B$, applying Proposition 4 to each with its own entry wave identifies the increment $h^{S_A}(s_A) - h^{S_B}(s_B)$ without any fresh cohort at t , under B4 for each cohort and comparability of the two cohorts' latent distributions at t : this is the constructive version of Corollary 7(ii), and it is what a generational refreshment leaves identifiable when B1 fails between entrants and incumbents. Second, if the two cohorts differ in a fixed attribute of the survey process that shifts every report additively by κ —the JLPS 2011 cohort's questionnaires are returned by mail while the 2007 cohort's are collected by interviewer visit—then no within-cohort adjustment removes κ , every level contrast identifies its estimand plus κ , and the diagnostics of Section 8, which compare designs within the same pair of cohorts, are unaffected. The combination $\vartheta := \{h^S(s) + \kappa\} - \{[h^{S_A}(s_A) - h^{S_B}(s_B)] + \kappa\}$ is free of κ but constrains only the *difference* of two conditioning contrasts: it equals $h^S(s)$ when the path has saturated between s_B and s_A and zero under a time-homogeneous linear path with $s_A - s_B = s - 1$; it does not identify either contrast separately. We record this because it is the situation of the JLPS and because the temptation to read ϑ as a level is strong.

8 Diagnostics for the Attrition Process

Because Proposition 2 and Proposition 4 identify the same quantity under different restrictions, their difference is an overidentification statistic, and by Proposition 5 its expectation is a combination of the bias terms; the same holds for Proposition 3 against Proposition 2.

Proposition 6 (Two directional diagnostics). *Define $\mathcal{T}_{NS} := \hat{h}_{SM} - \hat{h}_{EC}$ and $\mathcal{T}_{SD} := \hat{h}_{SSM} - \hat{h}_{SM}$. Add*

- **B5 (survivor-effect homogeneity).** $h^{S+} = h^S$: the mean conditioning shift is the

same among survivors of k and of $k + 1$ further waves.

Under B1–B5 both statistics have expectation zero. B5 is needed because $\mathcal{T}_{SD}$ compares estimands defined on different survivor sets: without it, $\mathbb{E}[\mathcal{T}_{SD}] = (h^{S+} - h^S) + \Delta_{SSM} - \Delta_{SM}$ even when B1–B4 hold, and a rejection can reflect a change in effect composition induced by the additional survival requirement rather than any failure of the attrition restrictions. The statistics are therefore compatibility checks among the maintained restrictions, not unique classifiers of which one fails. Under model (M) of Proposition 5: with stationary state-dependent attrition, $\mathbb{E}[\mathcal{T}_{NS}] = \Delta_{SM} - \eta = 0$ and $\mathbb{E}[\mathcal{T}_{SD}] = (h^{S+} - h^S) - \Delta_{SM}$; with non-stationary trait selection, $\mathbb{E}[\mathcal{T}_{NS}] = \Delta_{SM}$ and $\mathbb{E}[\mathcal{T}_{SD}] = (h^{S+} - h^S) + \Delta_{SSM} - \Delta_{SM}$, which need not vanish. In (M), then, $\mathcal{T}_{NS}$ has no power against stationary state dependence, and $\mathcal{T}_{SD}$ can respond to non-stationarity as well as to state dependence. Each statistic is a difference of subset means within two independent cohorts; its variance is estimated from influence functions that include the estimated group shares (Section 13), or by a person-level bootstrap within cohort that recomputes every arm and every share.

Proof. Immediate from Proposition 5: $\mathcal{T}_{NS}$ estimates $\Delta_{SM} - \eta$ and $\mathcal{T}_{SD}$ estimates $(h^{S+} - h^S) + \Delta_{SSM} - \Delta_{SM}$; substitute the values of $\Delta_{SM}, \Delta_{SSM}, \eta$ under (M)(i) and (M)(ii). $\square$

Two cautions on interpretation. A rejection indicates incompatibility among the maintained restrictions B1–B5; it does not uniquely identify the source of failure, and it does not say which design is wrong. A failure under which the compared designs move together—a mechanism that biases SM and EC alike, such as stationary state dependence in (M)—is invisible to $\mathcal{T}_{NS}$. The diagnostics are therefore complements to, not substitutes for, the overidentification tests of Theorem 4(c), which use a battery rather than a pair.

Monte Carlo evidence. Simulation 3 (Section 12) calibrates the designs to the size of the Japanese panel’s 2011 episode ($k = 4$, $n_{old} = 4,800$, $n_{new} = 960$, $R = 1,000$ replications per regime) under six attrition regimes generated within model (M). The constructive estimators have biases below 0.003 in absolute value (outcome units) and 95% coverage between 0.94 and 0.96 under the regimes their restrictions allow. Survival matching is biased under non-stationarity (−0.081, coverage 0.65) and under state dependence (+0.104, 0.53); the

entry-wave correction under state dependence $(+0.103, 0.36)$ but not under non-stationarity $(-0.002, 0.95)$; symmetric matching under non-stationarity $(-0.068, 0.76)$ but not under state dependence $(+0.002, 0.95)$, as Proposition 5(i)–(ii) implies for these processes. The diagnostics reject at rates between 0.039 and 0.056 under the three regimes in which B1–B4 hold; $\mathcal{T}_{\text{NS}}$ rejects at 0.62 under non-stationarity and at 0.040 under state dependence, as (M) implies, and $\mathcal{T}_{\text{SD}}$ at 0.99 under state dependence and at 0.08 under non-stationarity. Inverse-probability weighting on entry covariates, the design most users would reach for first, is unbiased only under selection on observables and is biased by -0.11 to -0.31 under every regime with selection on the unobserved trait, a reminder that the constructive designs’ restrictions concern the *timing* of selection, not its observability.

9 Design Theory: What a Panel’s Schedule Decides Before Data Arrive

The results so far take the panel’s schedule as given. This section treats the schedule as the object of choice. A design is the fielded support $\mathcal{S} \subset \{(e, t) : t \geq e\}$ —which entry cohorts are interviewed at which times—together with an assignment of instrument and mode to cells. Under the additive cell-mean structure of the companion identification paper, $\mu(e, t) = \alpha(t) + g(e) + \tau(s)$ with $\tau(1) = 0$ and $g(e_1) = 0$, stacking the fielded cells gives a linear system $\mu = X_{\mathcal{S}} \theta$ with $\theta = (\alpha, g, \tau)$ and $X_{\mathcal{S}}$ the design’s incidence matrix. Here $\mu(e, t)$ is the mean report of cohort e at t in the population, or among survivors after a selection correction such as (A2); with attrition, raw survivor means add selection terms β that vary by cohort and period and break the additive structure, which is the subject of Section 6. Two parameter vectors generate the same cell means iff their difference lies in $\mathcal{K} := \ker X_{\mathcal{S}}$; write $\mathcal{K}_{\tau}$ for the set of tenure components of its elements. Let d , the *stride* of the schedule, be the greatest common divisor of the spacings between entry cohorts. We use the *increment graph* of the companion identification paper: its vertices are the observed increments, the tenures u at which some cohort is observed at tenures u and $u + 1$ in consecutive periods, and u and u' are joined when two different cohorts are observed in the same two consecutive periods with

those increments. Condition C_d holds when every observed tenure is reached from tenure one through observed increments and the connected components of the increment graph are exactly the residue classes of the observed increments modulo d .

Theorem 5 (What a design identifies). *Let $\mathcal{S}$ be a fielded support with cell means following the additive structure and the normalizations $\tau(1) = 0$, $g(e_1) = 0$.*

- (i) (Matrix form.) *A linear functional $\lambda'\tau$ is identified from cell means if and only if λ annihilates $\mathcal{K}_\tau$. If, in addition, known linear restrictions $A\tau = a$ are imposed (anchors), the admissible directions are $\mathcal{K}_\tau \cap \ker A$: $\lambda'\tau$ is identified iff λ annihilates it, and τ is point identified on the observed tenures iff $\mathcal{K}_\tau \cap \ker A = \{0\}$. The rank of X_S , a basis of $\mathcal{K}_\tau$, and the identification of any proposed functional can be computed before fielding.*
- (ii) (Directions that are never identified.) *For every $m \in \mathbb{R}$ and every d -periodic ρ with $\rho(1) = 0$, the vector $h_\tau(s) = m(s-1) + \rho(s)$, $h_g(e) = m(e - e_1)$, $h_\alpha(t) = -m(t - e_1) - \rho(t - e_1 + 1)$ lies in $\mathcal{K}$. Hence no design identifies a level or the linear trend of τ from cell means alone, and when $d > 1$ no functional whose weights fail to sum to zero within some residue class modulo d other than that of $s = 1$ is identified; in particular, no ordinary second difference is.*
- (iii) (When those are all.) *If $\mathcal{S}$ satisfies C_d , the tenure components of the directions in (ii) span $\mathcal{K}_\tau$, and $\lambda'\tau$ is identified iff $\sum_s \lambda_s(s-1) = 0$ and $\sum_{s \equiv r} \lambda_s = 0$ for every residue $r \not\equiv 1 \pmod{d}$; the centred lag- d second differences $\Delta_d^2 \tau(s) = \tau(s+d) - 2\tau(s) + \tau(s-d)$ are then identified, and when $d = 1$ every ordinary second difference is. Without C_d , $\mathcal{K}_\tau$ can be larger even when $d = 1$ and all observed tenures are linked through tenures observed in common periods: with entries at waves 1, 3 and 4 observed through wave 4, the seven cells leave nine parameters with rank seven, and besides the affine direction the kernel contains $h_\tau = (0, 0, 1, 1)$ on tenures 1–4, $h_\alpha = (0, 0, -1, -1)$ on periods 1–4, and $h_g = (0, 1, 1)$ on cohorts 1, 3, 4, which moves $\tau(3) - 2\tau(2) + \tau(1)$. For staggered trapezoids $\{(e, t) : e \leq t \leq T\}$, the dimension of $\mathcal{K}_\tau$ equals the number of components of the increment graph; C_d holds whenever the last cohort is observed for $w := T - e_K \geq (e_2 - e_1) - d$ periods after its entry; with three cohorts the dimension*

is exactly $\max\{d, (e_2 - e_1) - w\}$; and with more cohorts that expression is an upper bound.

- (iv) (Mode assignment.) A mode change applied uniformly to all cells at a time t^* is absorbed in $\alpha(t^*)$ and costs no identification. A mode assignment that is a function of tenure—migrating long-tenure respondents to a cheaper mode first—adds a term $m(s)$ to the cell mean that loads on the same coordinates as $\tau(s)$; no functional separating m from τ is identified, and the mode effect is confounded with the conditioning path itself.

Proof.

- (i) Two parameter vectors generate the same cell means iff their difference lies in $\mathcal{K}$; with anchors, the admissible differences are the elements of $\mathcal{K}$ whose tenure components satisfy $Ah_\tau = 0$, and since A acts on τ alone their tenure projection is $\mathcal{K}_\tau \cap \ker A$. A linear functional is constant on an affine family iff it annihilates the family's direction space. (ii) At every fielded cell, $h_\alpha(t) + h_g(e) + h_\tau(s) = m[-(t - e_1) + (e - e_1) + (s - 1)] + [\rho(t - e + 1) - \rho(t - e_1 + 1)] = 0$, because $s - 1 = t - e$ and $e \equiv e_1 \pmod{d}$. The functional statements follow by taking $\rho = 0$, or $m = 0$ with ρ the indicator of one residue class $r \not\equiv 1$; for $d > 1$ the weights $(1, -2, 1)$ of an ordinary second difference at $s - 1, s, s + 1$ fall into more than one residue class, and at least one class other than that of tenure one has a nonzero sum. (iii) The spanning statement is Theorem 1(c) of the companion identification paper, and the trapezoid statements are its Lemma 1(o), (i), (ii) and (iv); the functional characterization is (i) with $\mathcal{K}_\tau$ spanned by $s - 1$ and the residue-class indicators. In the example the increment graph has the components $\{1, 3\}$ and $\{2\}$, and the rank and the null vector are verified by exact arithmetic (Section 12). (iv) A uniform switch adds a constant to all cells at t^* , which is a change in $\alpha(t^*)$. A tenure-dependent assignment adds $m(s_{it})$, a function of the same coordinate as $\tau(s_{it})$; the columns of $X_{\mathcal{S}}$ that carry $\tau(s)$ and $m(s)$ are identical, so their sum is the only identified object. $\square$

$\square$

The rank calculation turns design review into arithmetic: a proposed schedule is a matrix, and its identified space is computable before a single interview. Anchors enter the same calculation. A negative-control battery at a refreshment episode anchors the level of τ at that episode's tenure (Theorem 4); under C_d , a level anchor at a tenure congruent to one modulo d removes the affine direction and none of the periodic ones; a plateau restriction over $d+1$ consecutive tenures removes all d directions; an external estimate of the linear part of τ removes the affine direction only. The Japanese panel of Section 10 (entries at waves 1, 5 and 13, observed through wave 19) has 41 cells, 39 free parameters and rank 35; it satisfies C_4 , so the identified set of the path is four-dimensional. It identifies the eleven centred lag-4 second differences $\Delta_4^2\tau(s)$, $s = 5, \dots, 15$, and cross-class contrasts such as $\Delta^2\tau(2) - \Delta^2\tau(6)$ —a 14-dimensional space of functionals—but no ordinary second difference. Level anchors at tenures 5, 9 and 13, all congruent to one modulo four, as Theorem 4(a) would supply with an age-matched second refreshment, leave the three periodic directions, and so do the anchors the panel actually supplies, a level at tenure 5 and the increment $\tau(13) - \tau(9)$; a plateau over five consecutive tenures would leave none (Section 12).

9.1 The stride of a refreshment schedule

For staggered trapezoids, Lemma 1 of the companion identification paper gives a sufficient follow-up condition for the identified set of the path to have the minimal dimension d , and the exact dimension with three cohorts. Read as design rules these say three things.

Proposition 7 (Refreshment spacing as an identification decision). *Let a panel enter cohorts at waves $e_1 < \dots < e_K$ and be observed at every wave through T , with d the stride and $w = T - e_K$.*

- (i) (What the schedule can ever deliver.) *Without anchors, the identified set of the path contains $\tau + \{m(s-1) + \rho(s)\}$, ρ d -periodic with $\rho(1) = 0$: a schedule with stride $d > 1$ cannot distinguish, from cell means, a path from one that differs from it by a d -periodic pattern. Under C_d this is the whole identified set, and the identified curvature is spanned by the centred lag- d second differences and contrasts across residue classes.*
- (ii) (Coprime spacing.) *Adding a cohort at a wave e' with $\gcd(d, e' - e_1) = 1$ makes the*

stride one. If in addition the last cohort of the enlarged schedule is followed for at least $(e_2 - e_1) - 1$ periods after its entry, where e_2 is the second-earliest entry after the addition, the identified set is the affine line and every second difference of τ is identified.

(iii) (Follow-up.) With three cohorts, the identified set has dimension $\max\{d, (e_2 - e_1) - w\}$: each additional period of follow-up of the last cohort removes one dimension until the floor d is reached. With more cohorts, $w \geq (e_2 - e_1) - d$ remains sufficient for dimension d and $\max\{d, (e_2 - e_1) - w\}$ is an upper bound, but shorter follow-up of the last cohort can suffice: with entries at waves 1, 5, 6 and 9 and final wave 9, the last cohort has no follow-up and the identified set is nevertheless the affine line.

Proof.

(i) is Theorem 5(ii)–(iii). (ii): the stride after the addition is $\gcd(d, e' - e_1)$, because every $e_k - e_1$ is a multiple of d ; the dimension statement is Lemma 1(i) of the companion identification paper with stride one. (iii) is Lemma 1(ii) and (iv) there; the four-cohort example is verified by exact rank computation (Section 12), which also shows that among the four- and five-cohort trapezoids with first entry at wave 1, later entries in $\{2, \dots, 10\}$ and up to eight periods of follow-up, 176 attain dimension d with $w < (e_2 - e_1) - d$. $\square$

$\square$

Applied to the panel of Section 10: its entries at waves 1, 5 and 13 give $d = 4$. A fourth refreshment can restore stride one only if it enters at a wave e' with $\gcd(4, e' - 1) = 1$, that is, at an even wave: entry at wave 20 gives $\gcd(4, 19) = 1$ and entry at wave 22 gives $\gcd(4, 21) = 1$, whereas entry at wave 21 ($\gcd(4, 20) = 4$) leaves the stride at four and entry at wave 19 ($\gcd(4, 18) = 2$) halves it. Stride one is necessary for identifying ordinary curvature but not sufficient: with the entry at wave 20 the identified set has dimension 4, 3, 2 and 1 for final waves 20, 21, 22 and 23, so three periods of follow-up after the new entry are needed; here the sufficient condition of Proposition 7(ii), $w \geq (5 - 1) - 1 = 3$, is also necessary. Refreshment timing is usually decided on budgetary and coverage grounds; the result says it

is also an identification decision. The same arithmetic applies to any refreshed panel: a panel refreshed every two years on an annual interview calendar has $d = 2$ and identifies curvature only at lag two, and a panel refreshed at irregular intervals whose gaps have no common divisor can identify ordinary curvature once the follow-up is sufficient. Rotation designs with interrupted participation, such as the 4–8–4 pattern of the US Current Population Survey, are treated separately in the companion identification paper. Table 1 collects a few cases; every entry is an exact rank computation.

Table 1: Strides and identified-set dimensions of common refreshment schedules (staggered trapezoids observed at every wave through T), by exact rank computation. The last column applies when the dimension equals d .

Schedule (entry waves)		dimension of the identified set	follow-up needed for dimension d	identified curvature (dimension d)
annual	1	1	none	all $\Delta^2\tau$
refreshment:				
1, 2, 3, ...				
biennial	2	2	none	$\Delta_2^2\tau$ and
refreshment:			$(e_2 - e_1 = d)$	cross-class
1, 3, 5, ...				contrasts
Japanese panel:	4	4	none	$\Delta_4^2\tau$ and
1, 5, 13			$(e_2 - e_1 = d)$	cross-class
				contrasts
Japanese panel	1	4, 3, 2, 1 for	$T \geq 23$	all $\Delta^2\tau$ once
with a fourth		$T =$		$T \geq 23$
entry at wave 20		20, 21, 22, 23; 1		
		thereafter		
Japanese panel	4	4	none	$\Delta_4^2\tau$ and
with a fourth				cross-class
entry at wave 21				contrasts

Schedule (entry waves)	stride d	dimension of the identified set	follow-up needed for dimension d	identified curvature (dimension d)
1, 7, 10	3	$\max\{3, 6 - (T - 10)\}$	$T \geq 13$	$\Delta_3^2 \tau$ and cross-class contrasts once $T \geq 13$
a single refreshment at wave e_2	$e_2 - 1$	$e_2 - 1$	none	$\Delta_{e_2-1}^2 \tau$ and cross-class contrasts

The last row is worth pausing on: a panel with a single refreshment sample has stride equal to the refreshment lag, so it identifies curvature only at that lag, by comparing the entrants with the incumbents at tenures 1, $1 + (e_2 - 1)$ and $1 + 2(e_2 - 1)$. This is the design of most of the applied literature, and it is consistent with that literature’s reporting level contrasts under normalizations rather than shapes.

Theorem 6 (The balance–information trade-off). *Suppose aggregates are published as tenure-balanced averages: at every period the published statistic is $\bar{y}_t = \sum_s w_s \mu(t - s + 1, t)$ with weights $w_s \geq 0$, $\sum_s w_s = 1$, constant across t (one-level rotation balanced on time in sample). Then, under the additive structure with cohort effects absorbed into the period path,*

$$\bar{y}_t = \alpha(t) + \bar{\tau}, \quad \bar{\tau} := \sum_s w_s \tau(s),$$

so that (i) the tenure component of the published level is the constant $\bar{\tau}$: it does not vary over time, so changes in the published series are free of time-in-sample bias while its level carries the bias $\bar{\tau}$ in every period—balancing holds the bias constant, it does not eliminate it; and (ii) the published series depends on τ only through $\bar{\tau}$, which is not separately identified from the level of α , so no nonconstant functional of τ —in particular no normalization-free functional (differences, second differences, saturation tests)—is identified from the published

series alone. The information is lost when only the balanced aggregate is released, not by the balanced design as such: the rotation-group components from which the aggregate is formed identify what Theorem 5 says the corresponding staggered support identifies.

Proof. Substituting the additive structure into the balanced average gives $\bar{y}_t = \alpha(t) + \sum_s w_s g(t - s + 1) + \sum_s w_s \tau(s)$. With cohort effects absorbed into the period path (or under sampling equivalence $g \equiv 0$), the middle term is a function of t alone and joins α . The map $\tau \mapsto (\bar{y}_t)_t$ then depends on τ only through the scalar $\bar{\tau}$, which is (i); (ii) follows because a functional of τ is identified from the published series only if it is a function of the series, hence of $\bar{\tau}$, and because $\alpha(t) + \bar{\tau}$ does not separate its two summands. $\square$

Part (i) restates, in the present notation, a property the rotation literature established. Bailer (1975, 26, eq. (4.2)) shows that under a constant rotation group bias the level estimates are biased while month-to-month changes are not, and concludes that “neither estimator is ‘best’ for every statistic” (Bailer 1975, 30); Park, Kim, and Choi (2001, 1488, eq. (6)) establish the constant level bias and its cancellation in changes for balanced one-level rotation designs, and their component-level model (Park, Kim, and Choi 2001, 1489, eq. (7)) supports estimating rotation group bias from the rotation-group components. What (ii) adds is an explicit identification statement for the case in which only the fixed-weight aggregate is released—the series then depends on the time-in-sample path only through its weighted average, which is confounded with the period level—and its connection with the rank calculation of Theorem 5. It is the design-side counterpart of the absorption result in the companion identification paper, where the same orthogonality that protects two-way fixed-effects estimates from affine drift removes the affine drift from what those estimates can reveal. Releasing disclosure-approved rotation-group summaries alongside the balanced aggregate can preserve information that the aggregate loses. We leave the allocation problem—how to distribute a fixed budget of fresh interviews across injection times to maximize the information on the drift coefficient—open; numerical evaluation of the design information matrix on lattice designs suggests that two injections at maximal temporal spread dominate, but we have no proof and state no theorem.

10 Implementation: JLPS

The Japanese Life Course Panel Surveys (JLPS) provide the three-cohort configuration of Theorem 4, with a negative-control battery of 23 time-invariant childhood-circumstance items asked once at entry in every cohort. The results below are illustrations under the maintained restrictions (A1)–(A2) and B1–B5, not tests of them. The companion empirical paper implements the single-refreshment diagnosis at the 2011 episode (478 items) with the selection-reversal and Lee-trimming robustness checks whose population counterparts are Theorem 3 and Corollary 6. The birth years of the 2011 entrants lie within those of the incumbents (1966–1986): the birth-year supports overlap, and the analysis additionally assumes cohort comparability as specified in (A1)—equal latent distributions on the common support—which overlap does not establish and which differences in sampling frames, nonresponse at entry, mode (Section 7.4) or coding could violate. The battery consists of the 23 items whose question stems appear in all three cohorts’ entry questionnaires (the map from each item to its variable in the three entry questionnaires is built from the data provider’s variable labels; it is not redistributed with the replication materials and is available from the author to licensed users); equality of response categories and of administration across the entry questionnaires is assumed, not tested. The 2019 episode is the generational case of Corollary 7: entrants born 1987–1998 against incumbents born 1966–1986, an empty overlap, verified in the data. The estimand implemented at that episode is therefore the incumbent increment: the 2007 cohort (tenure 13, $n = 2,638$ at wave 13) against the 2011 cohort (tenure 9, $n = 619$) on their common birth-year support, with the same-pair negative-control contrast $\hat{\beta}_2^N(13) - \hat{\beta}_2^N(9)$ as the correction under (A2). Across the 23 items, stratified by three-year birth band and sex (Table 3), the median absolute standardized contrast is .033—the same order as the 2011 episode’s .031—with one rejection in 23 at a Benjamini–Hochberg $q < .10$ (the item recording a telephone, mobile telephones included, in the childhood home, whose availability depends on the era of childhood), nine equivalences by two one-sided tests with margin .10, and an inverse-variance pooled value of $-.035$. The standard error .0095 of that pooled value, computed as $(\sum_k 1/\text{se}_k^2)^{-1/2}$, treats the 23 item contrasts as independent; because the items are answered by the same respondents, we also computed a person-level

bootstrap within cohort (2,000 replications, every item contrast, stratum weight and pooling weight recomputed), which keeps their covariance: the bootstrap standard error is .0186, 2.0 times the independence value (a variance ratio, or design effect, of 3.84; the mean correlation between item contrasts across replications is .116), and the 95% percentile interval is $[-.072, .001]$, which includes zero. Without the rejected item the pooled value is $-.028$ (bootstrap standard error .0185). Under (A2) with a common loading Γ , increment estimates that skip the correction would be biased by Γ times the survivor-composition drift, estimated at about .035 Γ standard deviations in magnitude (.017 to .052 over $\Gamma \in [.5, 1.5]$). Because the interval for the pooled value includes zero, the battery does not establish that the survivor composition has drifted; what it limits, at the 95% level, is the size of the drift, and at the far end of the interval the implied bias would be .072 Γ (.036 to .108 over the same range of Γ). A small negative-control contrast measures the drift in survivor composition on the battery; it removes the target’s selection bias only to the extent that (A2) and its loading hold.

Practically, Theorem 1 converts into a one-line empirical check per item. For a continuous item, $\sup_y \hat{p} \hat{q}(y) / \hat{f}_2(y)$ over a trimmed support, with $\hat{q}$ the stayers’ reported density, is a plug-in estimate of $\Psi(0)$, the constraint ratio of Theorem 3 for the candidate of no conditioning (equal to $\text{ess sup } \bar{\pi}$ when the true shift is zero), and it must be read with Remark 2 in mind. A value above one excludes that candidate. A value well below one, together with a log-density that does not vary too fast, implies that shifts near zero belong to the location set, because $\Psi(h') \leq \Psi(0) e^{L|h'|}$ when $\log f_2$ is L -Lipschitz, the argument of Corollary 2(iii); a value near one does not imply a tight set, because $\Psi(0)$ says nothing about Ψ elsewhere. For a discrete item, the mass ratio $\max_a \hat{p} \hat{Q}(\{a\}) / \hat{F}_2(\{a\})$ of Corollary 4 needs neither trimming nor slack, and its population version is exact. We report this diagnostic alongside the item-level estimates in the companion empirical paper’s replication materials.

The constructive designs of Section 7 have been applied to the same panel at the 2011 episode, with the fresh cohort’s follow-up supplying $k = 4$ and $k + 1 = 5$ matched waves and the 2007 cohort’s entry wave supplying the entry-wave correction for the 268 of 478 items asked with the same coding at entry (computations in the R package `panelcond`; ag-

gregate outputs only). The arms are 2,797 continuing survivors of five consecutive waves against 963 fresh entrants, of whom 574 survive four further waves and 540 five. Item-level inference is by a joint person bootstrap with 500 replications that resamples persons within cohort and recomputes every arm and every group share, so it does not rely on the analytic formula of Section 13. Three findings preview the companion empirical paper. First, the designs disagree: across the 470 substantive items with variation, the sets of items flagged at a false-discovery rate of .10 number 28 (naive), 19 (survival matching), 13 (symmetric matching), and, on the 265 items with variation for which the entry wave is available, 22 and 21 (entry-wave correction, unadjusted and covariate-standardized); on that common set of 265 items the counts are 19, 15, 9, 22, and 21, with four items flagged by all five designs. Second, with bootstrap denominators the diagnostics reject at the 5% level for 13.1% ($\mathcal{T}_{\text{NS}}$) and 8.2% ($\mathcal{T}_{\text{SD}}$) of substantive items, and at 19% for $\mathcal{T}_{\text{SD}}$ on item nonresponse; given B5, the rejections are read as incompatibilities among the maintained restrictions rather than as counts of failures of a named kind. Third, the response-style margins that the companion empirical paper identifies as the locus of conditioning—a fall of .21 to .25 standard deviations in extreme-category use and a rise of .19 to .21 in midpoint use—are similar across all five designs, whereas the one substantive item flagged in earlier analyses of these data (employment, an entry-wave-corrected difference of 4.9 points) loses significance under covariate standardization and under the multiplicity of the full inventory. The sample analogue of the mass-domination criterion of Corollary 4, $\max_a \hat{p} \hat{Q}(\{a\}) / \hat{F}_2(\{a\})$ over the categories of each discrete item, with no trimming and no slack, exceeds one for seven of the 405 items with two to nine categories: party identification (1.96), intention to succeed to a family business (1.27), and five further items with ratios between 1.02 and 1.37, several of them rare-event items (parental death, receipt of public assistance). At the population level, for an item whose categories are all reported, the inequality holds under every attrition process, so a violation is evidence against the maintained model—refreshment validity, comparable coding, and deterministic monotone measurement—that no attrition process removes (Corollary 4). We nevertheless report the seven items only as *exploratory flags*, because the criterion is a maximum of estimated ratios whose sampling error is not calibrated here, because rare cells inflate the ratio (the denominator is the fresh cohort’s share of the category, estimated

from 963 entrants), and because no multiplicity adjustment is applied across 405 items. A one-sided confidence bound on the maximal ratio, with multiplicity control across items, is the appropriate inferential version and is left to future work.

11 Discussion

The results delimit a division of labour. What a refreshment sample cannot do by itself is separate conditioning from attrition selection when the location set is nondegenerate: Theorem 1 gives the exact residual ambiguity, which vanishes without attrition and, under outcome-independent attrition, whenever the refreshment density has tails thinner than exponential (Corollary 3). Within a cohort and without restrictions linking waves, the panel’s own history does not narrow it (Theorem 2), so, within the maintained model, additional identifying restrictions must exclude otherwise feasible completions. What a refreshment sample can do is threefold: bound the ambiguity by an observable density-ratio criterion—the Horowitz–Manski interval for an unrestricted map (Corollary 6), the location set of Theorem 3, the mass inequality for an item whose categories are all reported (Corollary 4); shrink it with covariates (Proposition 1) and with negative controls under a transport restriction; and, with a negative-control battery at each refreshment and the transport restriction (A2) with a known loading, remove it at the tenures observed at refreshment episodes. The no-conditioning restriction that the refreshment-sample attrition literature maintains (Hirano et al. 2001; Deng et al. 2013) selects one point of a set whose extent the data reveal, and whose plug-in width, Remark 2 warns, depends on where the analyst stops trusting the tails.

Three points bear on practice. First, the entry-wave functional of Das, Toepoel, and Soest (2011) can be read as an estimator of a survivor effect under a restriction on unconditioned latent outcomes (B4), or of a population effect under their stationarity assumption on conditioned outcomes. The two readings coincide under homogeneous effects and differ by the composition term $\mathbb{E}[\tau_i \mid S] - \mathbb{E}[\tau_i]$ otherwise (Simulation 3), so reports of entry-wave-corrected estimates should state which target they intend. Second, the diagnostics are compatibility checks among the maintained restrictions B1–B5, not classifiers. When non-stationary and

state-dependent attrition are both present, the estimator with the smallest bias in Simulation 3 is survival matching, whose restriction both failures violate, because the two failures bias it in opposite directions; and under stationary state dependence in model (M), $\mathcal{T}_{\text{NS}}$ has no power at all. A rejection is evidence that the set of restrictions is inconsistent, to be followed by the overidentification tests of Theorem 4(c). Third, the identified set of the conditioning path on a staggered trapezoid has dimension at least the stride of the refreshment schedule, and more when the follow-up of the last cohort is short; on irregular supports it can be larger or smaller, and the rank computation of Theorem 5, not a closed-form condition, is the general check.

The deterministic monotone-map model serves continuous items. For items whose categories are all reported it is a specification test (Corollary 4), and the response-style conditioning that is this programme’s main empirical finding is a stochastic partial merging of categories that the deterministic model does not represent. A distributional treatment of stochastic conditioning on ordered categories, extending Das, Toepoel, and Soest (2011) beyond the binary case, is not attempted here.

Our recommendations for panels with refreshment designs are therefore: report the identified set rather than assuming it away, stating the trim and slack of any plug-in version; report survivor and population targets separately when effects may be heterogeneous; design panels with at least two refreshments, an entry-wave negative-control battery, and where possible a *repeated tenure* across refreshment episodes, which in the location family adds a direct test of stationary selection; compute the rank of a proposed schedule before fielding it; choose refreshment waves whose spacings have no common divisor and follow the last refreshment cohort long enough (for the panel of Section 10, an entry at an even wave followed for at least three waves); and release disclosure-approved rotation-group summaries alongside balanced aggregates. These are identification decisions that are usually taken on other grounds.

12 Appendix: Simulation Evidence (ADEMP)

Three simulation studies, reported in the ADEMP format, examine the finite-sample content of the results, and three deterministic scripts verify the worked examples, the counterexamples, and the design-rank statements. Code and outputs are in the replication archive (`sims/sim1_bounds.R`, `sims/sim2_multicohort.R`, `sims/sim3_designs.R`, `sims/check_funnel_examples.R`, `sims/check_multiwave_completion.R`, `sims/check_design_rank.py`) no confidential data are involved. `sims/check_manuscript_values.R` asserts every simulation value quoted in the text and tables against a named cell of the output files, with the estimand and units of each. The data-generating processes and estimands below are those implemented in the archived scripts.

Simulation 1 (Theorem 1, Corollary 2, Theorem 3: a relaxed plug-in procedure).

Data-generating mechanism: bivariate Gaussian (Y_1, Y_2^*) with unit variances and $\rho = .5$; logistic outcome selection $\pi = \Lambda(a + \beta Y_2^*)$ with $\beta \in \{0, .3, .6\}$ and a tuned to retention $p \in \{.6, .8\}$; location conditioning $h \in \{0, .2\}$; $n = 4,000$ plus a refreshment sample of 2,000; 200 replications per cell (in two cells one replication produced an empty plug-in set and is excluded from the set summaries). *Estimand:* h . *Methods:* a relaxed plug-in set $\{h' : \sup_y \hat{p} \hat{q}(y + h') / \hat{f}_2(y) \leq 1 + \kappa\}$ with kernel densities, the supremum taken over the support trimmed at 10% of the refreshment density's mode, slack $\kappa = .10$, and candidates $h' \in [-1, 1.5]$ on a grid of step .01 that contains the true values; the naive comparator is quantile alignment ignoring attrition. As population benchmarks the script computes, without sampling, the identified set of Theorem 3 for the same process, $h + [0, \Delta^*]$ (a point when $\beta = 0$), and the population analogue of the relaxed procedure (the same trim and slack applied to the true densities). *Performance:* The naive estimator's bias is essentially zero at $\beta = 0$ (at most .004 in absolute value), .056–.058 at $\beta = .3$ and .114–.116 at $\beta = .6$ when $p = .8$, and .116–.118 and .218–.220 when $p = .6$: raising retention from .6 to .8 halves it. The population identified sets have widths 0, .145 and .262 at $p = .8$ and 0, .260 and .489 at $p = .6$ for $\beta = 0, .3, .6$. The relaxed plug-in sets are wider—mean widths .25–.32 at $p = .8$ and .58–.68 at $p = .6$ —and positive at $\beta = 0$, where the identified set is a point. Their widths track those of the population analogue of the relaxed procedure (.31–.38 at $p = .8$,

.61–.71 at $p = .6$), so they are properties of the trim and the slack, not of the identified set (Remark 2), and Figure 1 reports them as such. Coverage of the true h by the relaxed plug-in set, counting an endpoint equal to the truth as covering it, is .930–1 at $p = .6$ and .760–.965 at $p = .8$, falling with β ; a uniformly valid version requires inference-aware calibration of the slack, in the spirit of intersection-bounds methods, which we leave to future work.

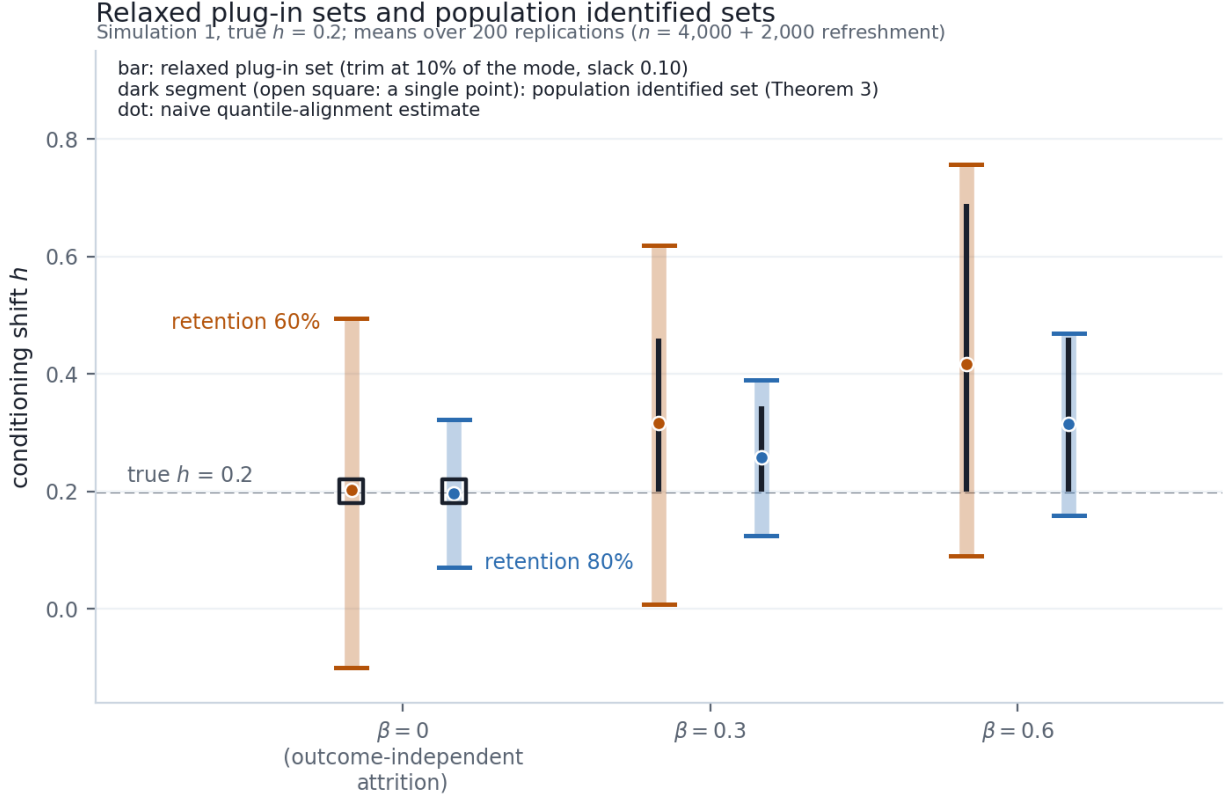


Figure 1: Relaxed plug-in sets and population identified sets, Simulation 1, true $h = 0.2$. Bars: mean over 200 replications of the relaxed plug-in set (support trimmed at 10% of the refreshment density’s mode, slack 0.10). Dark segments: the population identified set of Theorem 3 for the same data-generating process (open squares: a single point). Dots: mean naive quantile-alignment estimate. The plug-in widths reflect the trim and the slack; the population sets widen with selection strength and narrow with retention.

Simulation 2 (Theorem 4, Corollary 7). *Data-generating mechanism:* three cohorts entering at waves 1, 5 and 13 ($n = 3,000$ each), compared at wave 13; a latent trait $u_i \sim N(0, 1)$ and per-wave retention $\Lambda(1.6 + 0.6u_i)$, so that the first cohort has passed twelve response decisions and the second eight; target item $y = g_e + \tau(s) + u + e$ with $e \sim N(0, 1)$ and $\tau(s) = .15(1 - e^{-(s-1)/4})$; a $K = 23$ negative-control battery measured at entry, $x_k = .6u + \eta_k$;

a generational variant adds a cohort effect $g = .35$ to the wave-13 entrants' target item; a contaminated variant adds recall drift .15 to three battery items of the first cohort; 300 replications. *Estimands*, in units of the target's population standard deviation $\sqrt{2}$: the incumbent increment $(\tau(13) - \tau(9))/\sqrt{2} = .15(e^{-2} - e^{-3})/\sqrt{2} = .0091$ and the level contrast $(\tau(13) - \tau(1))/\sqrt{2} = .1008$. *Methods*: contrasts of survivors' raw means, divided by $\sqrt{2}$; negative-control corrections subtract Γ times the mean raw battery contrast of the same pair of cohorts, with the true raw loading $\Gamma = 1/.6$ (exact (A2)) or a unit loading on the population-standardized scale ($\Gamma = \sqrt{2}/\sqrt{1.36}$, an under-correction by the factor .73); a screened correction removes items flagged by a per-item outlier screen ($|z| > 2.5$, centred at the median item contrast) before pooling by the mean, and a median-pooled correction uses the median item contrast. The script also computes the population limit of each estimator by numerical integration. *Performance*: the naive increment is biased by .152 (population limit .151) from differential selection; the correction with the true loading has bias .002 (Monte Carlo standard error .002; population limit zero) and RMSE .042; the unit loading leaves .043 (limit .041)—a reminder that (A2) is a substantive assumption whose miscalibration under-corrects proportionally. The corrected level contrast has bias .003 (standard error .002) in the age-matched design and $-.250$ in the generational design, where the corrected contrast contains $-g/\sqrt{2} = -.247$; the incumbent increment stays unbiased there ($-.000$). The battery-homogeneity Q test rejects at .003–.007 at nominal .05 without drift and at .34 with three drifted items; the screen flags at least one of the three drifted items in 68% of replications, against 2% for the same three items without drift, and flags an undrifted item at a per-item rate below .01. Undetected drift contaminates the mean-pooled correction (bias $-.017$); screening before pooling ($-.007$) and median pooling ($-.006$) reduce it.

Simulation 3 (constructive designs and diagnostics; Section 7–Section 8). *Data-generating mechanism*: model (M) of Proposition 5 with a continuing cohort of $n_{\text{old}} = 4,800$ entering at wave 1 and a fresh cohort of $n_{\text{new}} = 960$ entering at wave 5 (dose $k = 4$); latent outcome $Y_{it}^* = 0.5X_i + 0.5U_i + \varepsilon_{it}$ with $X_i \sim \text{Bernoulli}(.5)$, $U_i \sim N(0, 1)$, $\varepsilon_{it} \sim N(0, 1)$ independent across waves, untreated standard deviation 1.146; a heterogeneous conditioning shift $\tau_i(s) = 0.30\{1 - e^{-(s-1)/4}\}(1 + \gamma U_i)$ whose population mean at $s = 5$ is 0.190 (0.166 standard deviations); and a dropout logit $-2.2 + a_X X_i + b_w U_i + a_E \varepsilon_{iw}$ for the decision

taken at wave w . Six regimes: MCAR; MAR on X ($a_X = .8$); MNAR on the trait ($a_X = .4$, $b_w = .8$); non-stationary MNAR on the trait, in which the trait coefficient is 1.4 at the continuing cohort’s waves before the comparison wave and .8 at every other wave of both cohorts, so that trait selection is *stronger* in the continuing cohort’s selection period; state-dependent MNAR ($a_X = .4$, $b_w = .4$, $a_E = .8$); and both failures together. Heterogeneity is $\gamma = .3$ (survivor targets h^S and h^{S+} differ from the population shift and from each other, the case B5 excludes) or $\gamma = 0$. *Estimands*: h^S (SM, EC, naive), h^{S+} (SSM), the population shift (IPW). *Methods*: the five estimators with influence-function standard errors that include the estimated group shares (Section 13); the diagnostics $\mathcal{T}_{NS}$ and $\mathcal{T}_{SD}$ at the 5% level. *Performance measures*: bias against each estimator’s own estimand and against the population shift, coverage of 95% intervals, rejection rates. $R = 1,000$ replications per regime and setting. The two settings of γ within a regime use the same seed, so their data are identical apart from the shift (common random numbers) and paired differences isolate the effect of heterogeneity. Part D adds the variance experiments described below. Code: `sims/sim3_designs.R`.

Table 2: Simulation 3, bias against each estimator’s own estimand in outcome units (95% coverage in parentheses) and rejection rates of the two diagnostics at the 5% level; 1,000 replications per regime. Monte Carlo standard errors of the biases are at most 0.002 and of the rejection rates at most 0.016.

Regime							
$(\gamma = 0.3)$	naive	SM	SSM	EC	IPW	$\mathcal{T}_{NS}$ rej.	$\mathcal{T}_{SD}$ rej.
MCAR	+ .001 (.94)	− .000 (.95)	− .000 (.95)	+ .001 (.95)	.000	.046	.053
MAR on X	− .058 (.75)	− .001 (.96)	− .002 (.95)	− .001 (.96)	− .001	.041	.056
MNAR trait	− .198 (.00)	.000 (.94)	+ .002 (.94)	− .001 (.94)	− .198	.039	.056
MNAR non- stationary	− .279 (.00)	− .081 (.65)	− .068 (.76)	− .002 (.95)	− .292	.623	.078

Regime							
$(\gamma = 0.3)$	naive	SM	SSM	EC	IPW	$\mathcal{T}_{\text{NS}}$ rej.	$\mathcal{T}_{\text{SD}}$ rej.
MNAR	−.127	+.104	+.002	+.103	−.115	.040	.993
state- dependent	(.17)	(.53)	(.95)	(.36)			
MNAR	−.292	+.012	−.062	+.078	−.306	.430	.945
both (non- stationary and state- dependent)	(.00)	(.93)	(.80)	(.59)			

The first five rows of Table 2 match Proposition 5(i)–(ii) row by row: survival matching fails under non-stationarity and under state dependence, symmetric matching only under non-stationarity, the entry-wave correction only under state dependence, and under state dependence survival matching and the entry-wave correction carry the same bias (+.104 and +.103), so that $\mathcal{T}_{\text{NS}}$ keeps its nominal size there. The sixth row is a case the diagnostics cannot classify: with both failures present, $\mathcal{T}_{\text{SD}}$ rejects almost always and $\mathcal{T}_{\text{NS}}$ less than half the time, and the estimator with the smallest bias is survival matching—not because its restriction holds but because the two failures bias it in opposite directions. A rejection is evidence of incompatibility among the maintained restrictions, and a non-rejection of one statistic is not evidence that the corresponding restriction holds.

Survivor versus population targets (Proposition 4). With heterogeneous shifts ($\gamma = 0.3$), dropout increases with U , so the survivors carry *smaller* shifts than the population: under MNAR on the trait the survivor target averages .170 against a population shift of .190. The entry-wave correction is unbiased for the survivor effect h^S (−.001) and biased for the population shift (−.021, or 11% of $\tau(5)$); with homogeneous shifts ($\gamma = 0$) the two targets coincide and the population bias is −.001. Because the two settings share their data, the

paired difference of the population biases, $-.020$, is the composition term $\mathbb{E}[\tau_i | S] - \mathbb{E}[\tau_i]$ up to a Monte Carlo error below $.0001$. Read as an estimate of the population effect under a stationarity-of-attrition-bias assumption on the *conditioned* outcome, the functional is off by that composition term whenever survival selects on the heterogeneity; read as an estimate of h^S under B4, it is unbiased here. Survival matching and symmetric matching show the same gap of about $.02$ between their survivor and population readings. The paired comparison also shows the role of B5: the $\mathcal{T}_{SD}$ rejection rate under MNAR on the trait is $.056$ with heterogeneous shifts and $.044$ when the composition term $h^{S+} - h^S$ is switched off (paired difference $.012$, Monte Carlo standard error $.005$).

Variance (Section 13). In Part A the share terms are small relative to the within-group variances, and the coverage of the entry-wave correction is essentially the same with and without them (for example $.945$ and $.944$ under MNAR on the trait). Part D shows when they matter. In D1, a null experiment with well-separated groups (entry and comparison responses both equal to $U + N(0, .1^2)$ with $U \sim \text{Bernoulli}(.5)$, survival $S = U$ in both cohorts, 500 entrants per cohort, no conditioning, 5,000 replications), the influence-function variance of the entry-wave correction averages $.00104$, against an empirical variance of $.00106$ and a theoretical value of $.00104$, and gives coverage $.947$ and a $\mathcal{T}_{NS}$ rejection rate of $.056$; the within-group formula without share terms averages $.00054$ and gives $.839$ and $.706$. In D2, an exact counterexample ($E = S$, $Y = 1$ among survivors, a constant fresh outcome), the entry-wave estimate equals the survival share $\hat{p}$, whose variance is $p(1-p)/n$; the influence-function estimate is its plug-in version $\hat{p}(1-\hat{p})/n$, while the within-group formula returns zero. In D3, a person-level bootstrap within cohort that recomputes every arm and every share (200 replications, 200 resamples each) agrees with the influence-function standard errors to within one per cent in D1 and in the MNAR-trait and MNAR-state regimes.

Deterministic checks (no sampling). `check_funnel_examples.R` reproduces, on a fine grid with stated tolerances, the identified set $\{-1\}$ for $f_2 = N(0, 1)$, $Q = N(-1, 1)$, $p = .5$; the attainment of $\Delta^* = \beta\sigma^2$ in the Gaussian-logistic example at $\alpha = -1$; the point $\{0\}$ under outcome-independent attrition with Gaussian tails; the Laplace and logistic half-widths $\log(1/p)$ and the Cauchy half-widths $(1-p)/\sqrt{p}$; the central-binding example after

Corollary 2 at $\sigma = .5$ and $.8$; the disconnected identified sets of the modulated Cauchy and logistic densities of Corollary 3, together with the exponential-tail outer bound and the failure of the closed form without the Lipschitz condition; the Lipschitz neighbourhood of Corollary 3(iii); the discrete examples of Corollary 4; the general-map example of Corollary 5; the retention-only covariate of Proposition 1; and a case in which the per-pair outer bound of Theorem 4(a) is strictly larger than the joint set (a Laplace episode with $C_1 = 0$, $N_1 = 1$ and a Gaussian episode with $C_2 = 10$, $N_2 = 1$, both at $p = .5$, with $\Gamma \in [0, 1]$: the Gaussian episode forces $\Gamma = 0$, so the joint set for the first episode is $\{0\}$ rather than $[-\log 2, 0]$). `check_multiwave_completion.R` enumerates, without sampling, a four-wave cohort on the latent lattice $\{-5, \dots, 5\}$ with a binary entry-only negative control, all eight response patterns (interrupted ones included), true conditioning by integer location shifts $(0, 1, 0, 1)$, and a selection kernel that depends on the whole latent path including the never-reported coordinates and that excludes the lowest latent value from responding at wave 4. For every candidate map $c'_4 = c_4 + \epsilon$, $\epsilon \in \{-3, \dots, 3\}$, with one arbitrary choice of candidate maps at the intermediate waves (shifts of $+1$ and -1 at waves 2 and 3), the completion of Lemma 2 is constructed explicitly. Exactly one non-trivial candidate, $\epsilon = +1$, satisfies the wave-4 domination inequality; for it and for $\epsilon = 0$ every pattern sub-measure, the entry marginal including the negative control, and the refreshment marginal are reproduced within the script's tolerance of 10^{-10} , the kernel is a valid Markov kernel, the survivors' latent mean at wave 4 moves by $-\epsilon$, and the negative-control contrast does not move; for every other ϵ the leftover L_4 has a negative entry. Restricting the same example to monotone patterns verifies the sequential form $\pi'_j = dN_{j+1}/dN_j$. The script also enumerates all 35 weakly increasing maps on four categories and confirms in every instance that the mass criterion Equation 1 coincides with a linear-programming feasibility test, with the identity as the sole feasible map when all categories are reported and non-identity maps feasible only when a category goes unreported; and it confirms the Gaussian widths of Corollary 6 (.390, .700, 1.288, 1.596 at $p = .9, .8, .6, .5$) by closed form, numerical integration, and a linear programme on a grid. `check_design_rank.py` computes, in exact rational arithmetic, the rank and the projected kernel of every design quoted in Section 9 and Table 1, the example of Theorem 5(iii) and its null vector, the four-cohort example of Proposition 7(iii), the effect of level and plateau

anchors on the Japanese panel’s support, and, on all 2,214 staggered trapezoids with three to five cohorts, first entry at wave 1, later entries in $\{2, \dots, 10\}$ and up to eight periods of follow-up, the component count, the sufficient follow-up condition, the exact three-cohort formula and the upper bound of the companion identification paper’s Lemma 1. On 65 of these designs every pair of observed tenures is linked through tenures observed in common periods although the identified set is larger than d .

13 Appendix: Variance of the Design Estimators

Each estimator of Section 7 and each diagnostic of Section 8 is a difference of subset means computed within two independent cohorts. For a cohort of n persons and a subset indicator B_i , the mean $\bar{Y}_B = \sum_i B_i Y_i / \sum_i B_i$ has influence function $\psi_i = B_i(Y_i - \mu_B) / \Pr(B)$; a linear combination of subset means within the cohort has the corresponding combination of influence functions, and its variance is estimated by $\sum_i \hat{\psi}_i^2 / n^2$. For the entry-wave correction, with S survival through t , E the entry response and Y the response at t , the continuing-cohort component $\hat{A} = \bar{Y}_S - (\bar{E}_S - \bar{E})$ satisfies

$$n \text{Var}(\hat{A}) \approx \frac{\text{Var}\{Y - (1-p)E \mid S\}}{p} + (1-p) \text{Var}(E \mid \text{NS}) + p(1-p)(\mu_{E,S} - \mu_{E,\text{NS}})^2,$$

where the last term comes from estimating the survivors’ share p ; a variance computed conditionally on the group sizes omits it. The diagnostics $\mathcal{T}_{\text{NS}} = (\bar{E}_S - \bar{E}) - (\bar{Y}_m - \bar{Y})$ and $\mathcal{T}_{\text{SD}} = (\bar{Y}_{S+} - \bar{Y}_S) - (\bar{Y}_{m+1} - \bar{Y}_m)$, where $\bar{Y}_{S+}$ averages the continuing survivors who also respond at $t+1$ and $\bar{Y}_m, \bar{Y}_{m+1}$ the fresh respondents who go on to respond for k and $k+1$ further waves, contain the analogous share terms in both cohorts. A person-level bootstrap within cohort that recomputes every arm and every share targets the same variance. Simulation 3 and version 0.1.4 of the R package `panelcond` use the influence-function formulas; the empirical results of Section 10 use the bootstrap.

14 Appendix: The Negative-Control Battery at the 2019 Episode

Table 3: Negative-control contrasts between the 2007 cohort at tenure 13 and the 2011 cohort at tenure 9, on their common birth-year support (1966–1986), standardized and pooled over three-year birth band by sex strata; aggregates computed from the licensed JLPS microdata by the replication kit. ^a Rejected at a Benjamini–Hochberg $q < .10$. ^b Standard error under independence across items; the person-bootstrap standard error is .0186 (design effect 3.84).

Item (reported once at entry)	d	SE
Possessions in the home at age 15: air conditioner	−.033	.045
television	−.032	.047
personal computer or word processor	.015	.045
video recorder	.040	.045
piano	−.074	.045
radio	−.092	.045
refrigerator	−.103	.046
second home	.009	.049
study desk	−.033	.045
reception room furniture	−.051	.045
owner-occupied home	−.014	.045
literature collection or illustrated encyclopedia	−.012	.045
farmland or forest (excluding a kitchen garden)	−.015	.045
encyclopedia	.025	.045
works of art or antiques	.023	.045
car	.022	.045
microwave oven	−.020	.045
telephone (mobile telephones included) ^a	−.176	.045
bath	−.069	.046
Number of books in the home at age 15	.078	.049
Family atmosphere at age 15	−.106	.045

Item (reported once at entry)	d	SE
Household living standard at age 15	−.096	.045
School grades in junior high school	−.069	.046
Inverse-variance pooled value	−.035	.0095 ^b

Data and code availability

The simulations and deterministic checks generate their own data; no restricted data are needed to reproduce the appendix. Their code and outputs, the figure script, and the manuscript source are in the replication archive at <https://github.com/sokubo/paper-refreshment-designs-replication>; the version corresponding to this manuscript is tag `paper-v0.5` (computational commit `0869ad8`). The archive’s README maps each table, figure and quoted simulation value to the script and output file that produce it, and `sims/check_manuscript_values.R` asserts those values against named output cells. The empirical illustration of Section 10 uses the integrated release (waves 1–19) of the Japanese Life Course Panel Surveys (JLPS) distributed to participants of the survey project; the microdata cannot be redistributed. Other researchers can apply for JLPS releases to the SSJDA Data Archive of the Center for Social Research and Data Archives, Institute of Social Science, The University of Tokyo; those releases differ from the integrated file, so an analysis of them with the archived code is a re-analysis with the same design and definitions rather than a replication. The archive contains the code that reads the licensed files and produces every aggregate quoted in Section 10 (`kit/R/`), a document stating the access route, the variables and the recoding (`kit/KIT_README.md`), and a synthetic data set with the outputs the code produces on it, so that the workflow can be run end to end without the licensed data. The aggregate outputs computed from the licensed data are reported in the paper (Section 10, Table 3) and are not otherwise redistributed. The R package `panelcond`, version 0.1.4 (<https://github.com/sokubo/panelcond>, tag `v0.1.4`), implements the estimators, the diagnostics, the variance formulas of Section 13 and the design-rank calculation.

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